



Outline

1. Overview
2. The Simulation Environment
 - 2.1 The concept of Sub-systems
3. Component Library
4. Implemented Modeling
5. Parameters' Scanning (*Sweep*)
6. "Optimization" Routines
7. Vizualizers: Electrical and Optical
8. Samples: WDM Systems, SCM Systems
9. Final Remarks



OptiSystem

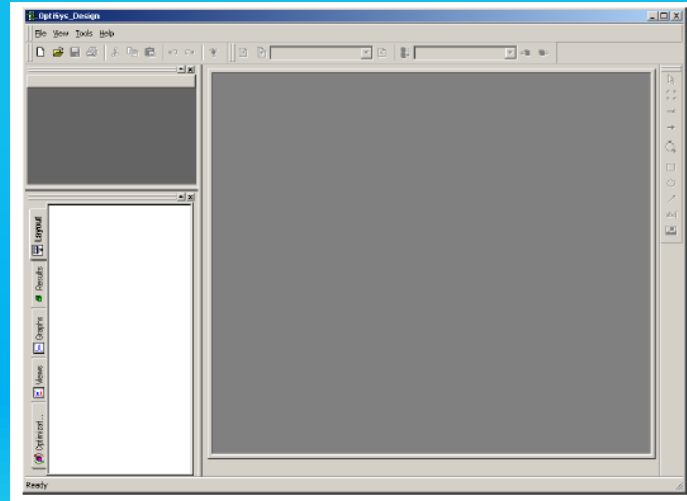
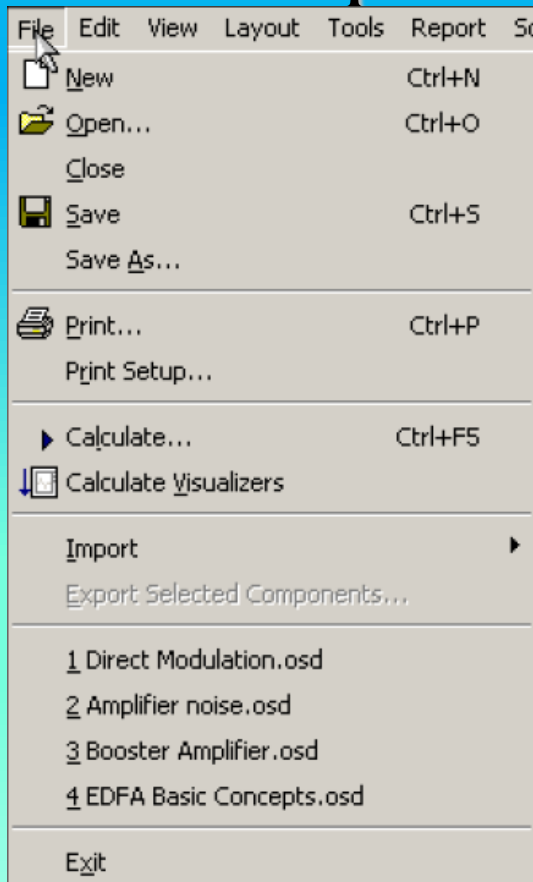
Software product for the design of optical fiber links



1. Overview

- Tool for the design&analysis of optical fiber links
- Applications: FTTH, projects of CATV, PON, rings SONET/SDH, optical amplifiers, WDM or TDM, SCM
- Component library, all *package*
- Components and Visualizers: Optical and Electrical
- Modeling: dynamic and stationary conditions
- Frequency (λ) and time domain (t)
- Co-simulation: MatLab, EDA (i.e. Spice), software Optiwave
- Cost estimation of the simulated layout

2. The simulation workspace



- *Project Overview*
- *Component Library*
- *Project Browser*
- *Parameter Groups*
- *Window showing the calculation evolution*



Workspace

Multiple Layouts
Language: Script

Project Overview

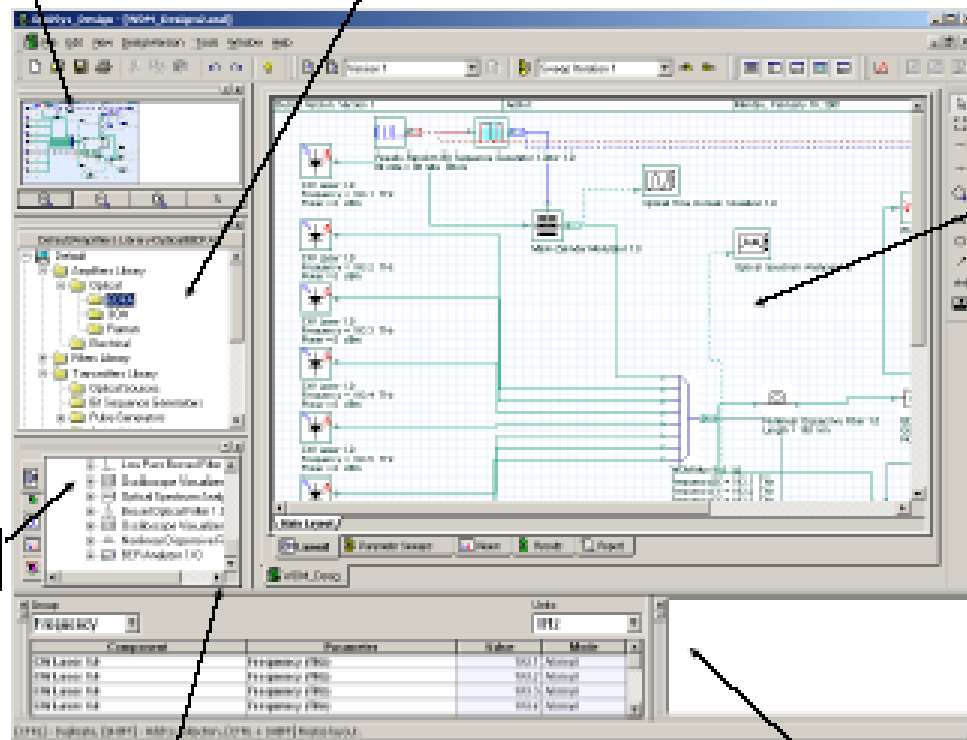
Component Library

Layout Editor

Project Browser

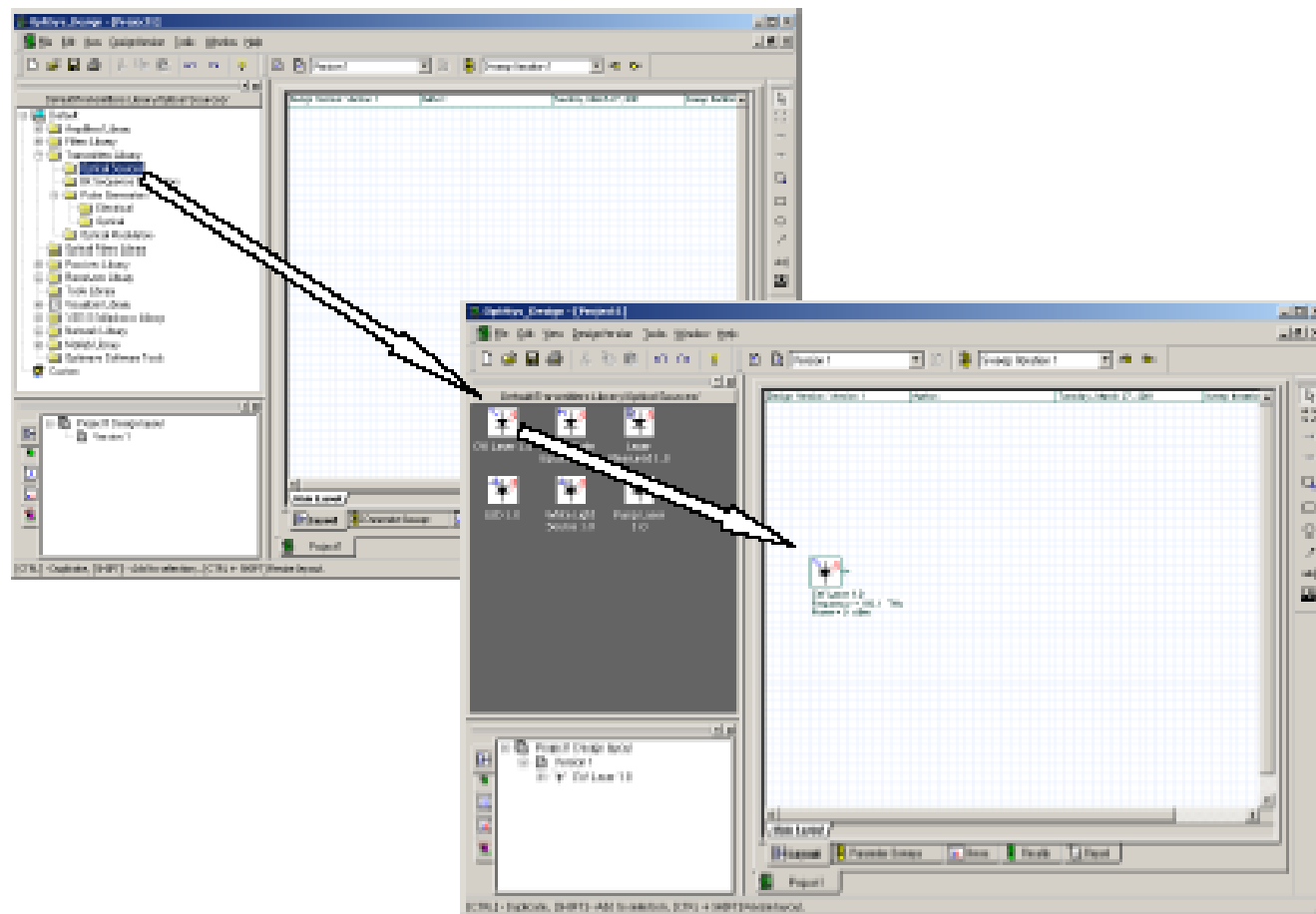
Parameter Groups

Output Window



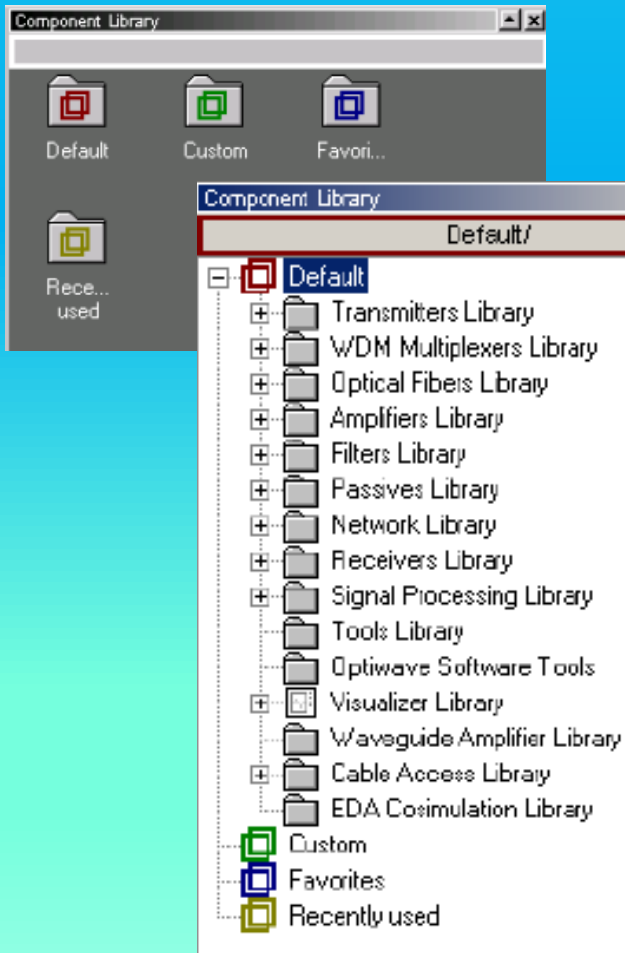


How to select components in the Component Library





Elements in the Layout



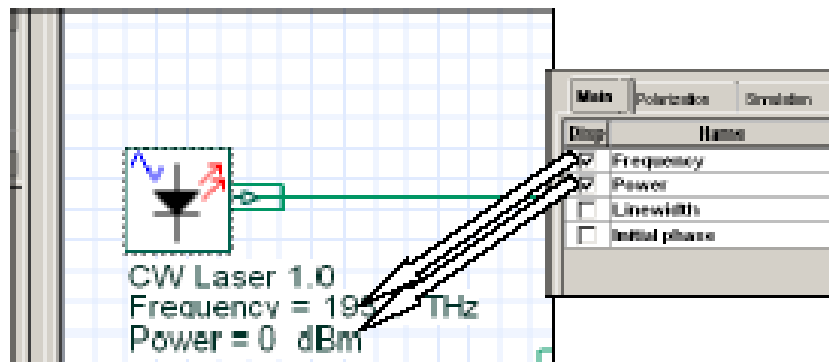
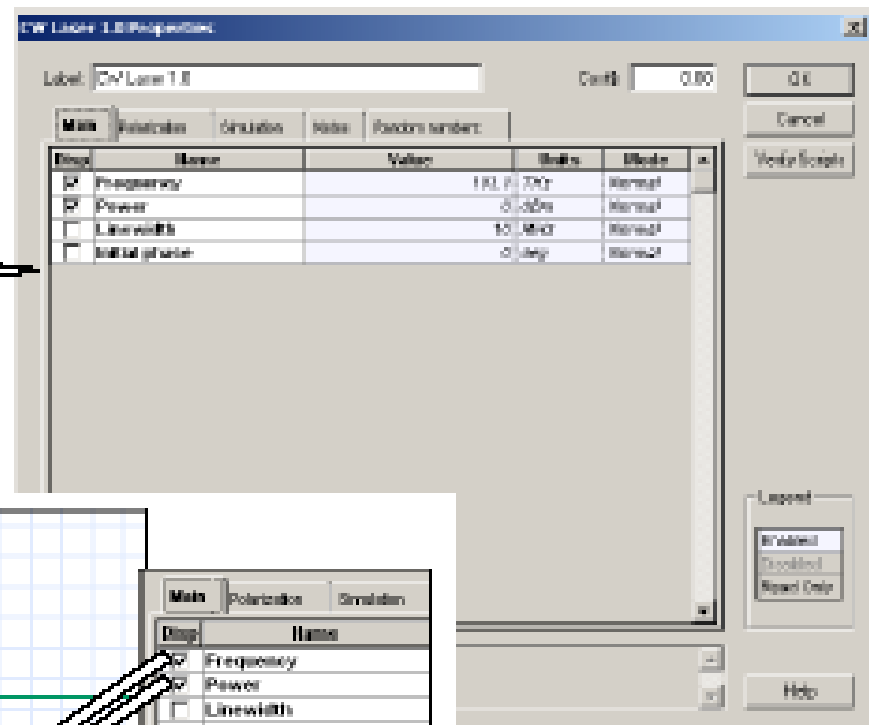
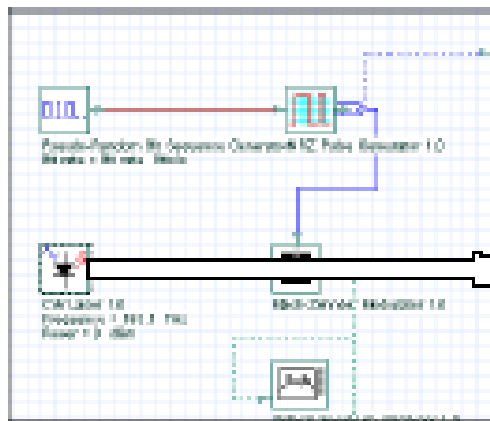
Project Browser

Default

Name	Value
Version 1	10
CW Laser 1.0	CW Laser 1.0
Ports	
Output	Optical Signal
Parameters	
Low Pass Bessel Filter 1.0	Low Pass Bessel Filter 1.0
Mach-Zehnder Modulator...	Mach-Zehnder Modulator 1.0
NRZ Pulse Generator 1.0	NRZ Pulse Generator 1.0
Optical Attenuator 1.0	Optical Attenuator 1.0
Optical Spectrum Analyze...	Optical Spectrum Analyzer 1.0
Optical Time Domain Visu...	Optical Time Domain Visualizer 1.0
Oscilloscope Visualizer 1.0	Oscilloscope Visualizer 1.0
Photodetector PIN 1.0	Photodetector PIN 1.0
Pseudo-Random Bit Seq...	Pseudo-Random Bit Sequence Generator 1.0



Input Parameters of each Component





Additional Functionalities

Frequency Units

Disp	Name	Value	Units	Mode
☒	Frequency	183.1	THz	Normal
☒	Power	0	dBm	Normal
☐	Linewidth	50	THz	Normal
☐	Initial phase	0	deg	Normal

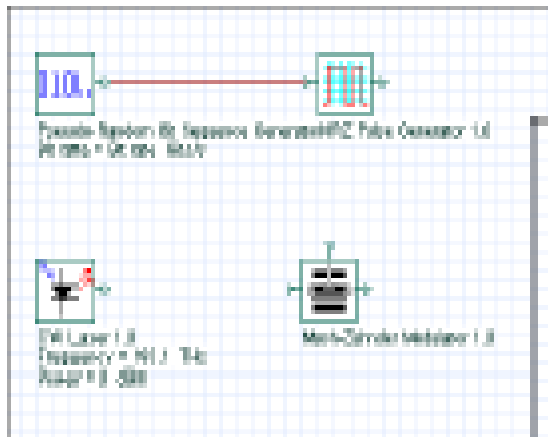
Power Units

Disp	Name	Value	Units	Mode
☒	Frequency	183.1	THz	Normal
☒	Power	0	dBm	Normal
☐	Linewidth	50	THz	Normal
☐	Initial phase	0	deg	Normal

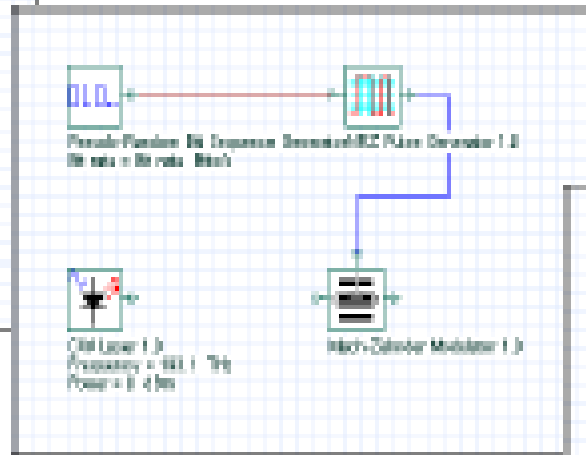


How to Set an Optical Transmitter

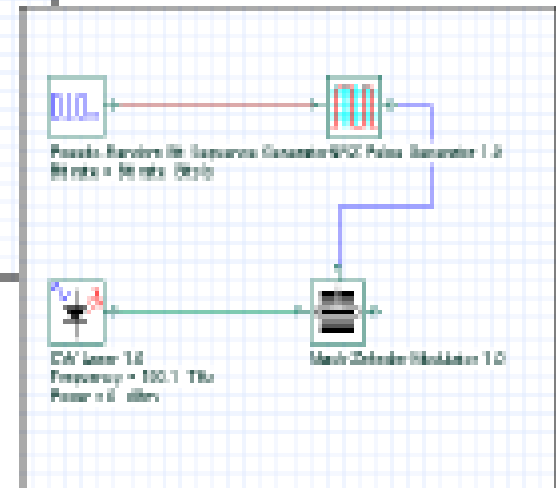
1



2



3

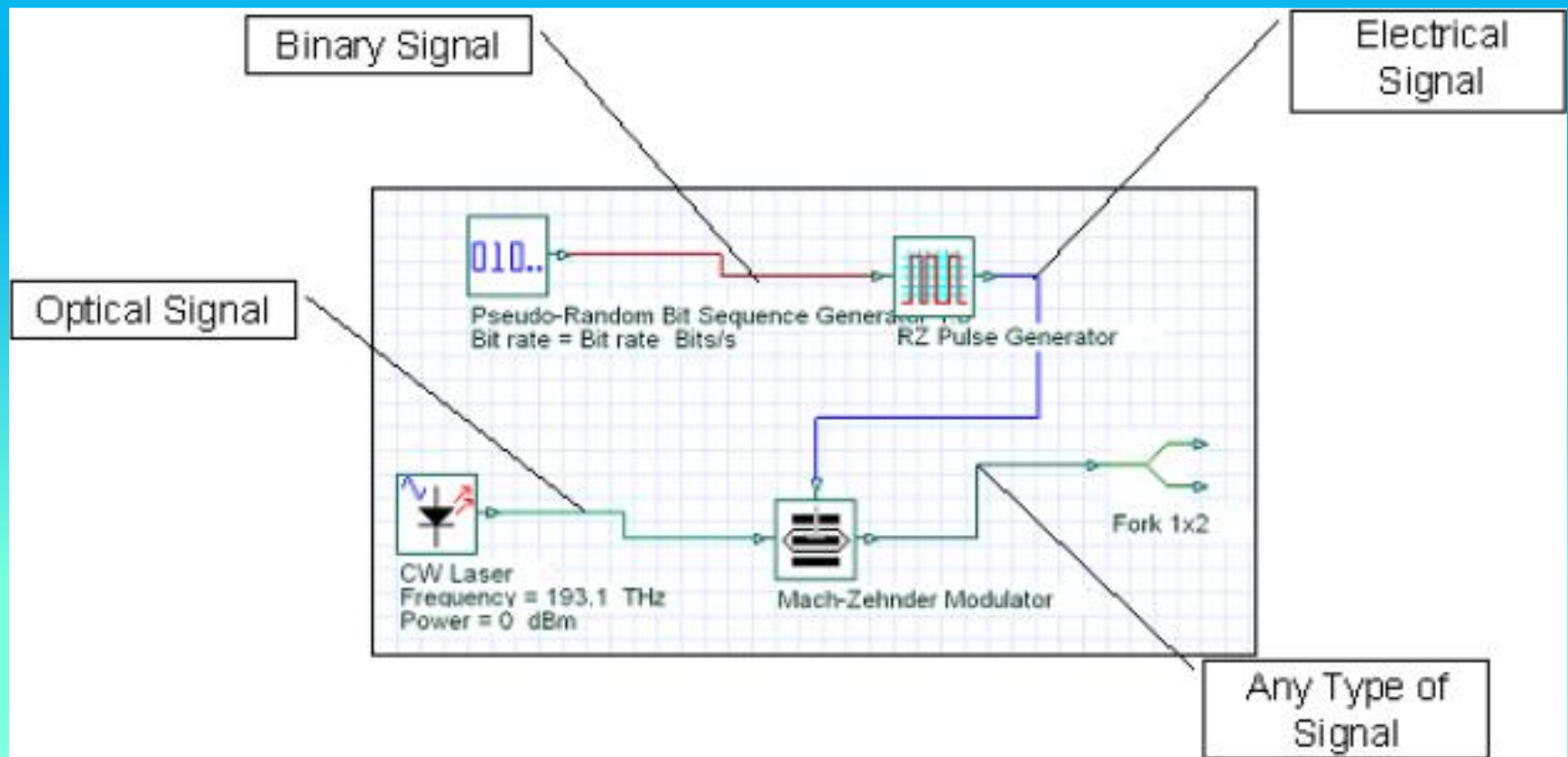


Connecting components





Signals and Connections





Exercise

1a. Set an optical transmitter operating at 1556 nm and calculate!

1b. Visualize the results (OSA, Oscilloscope)

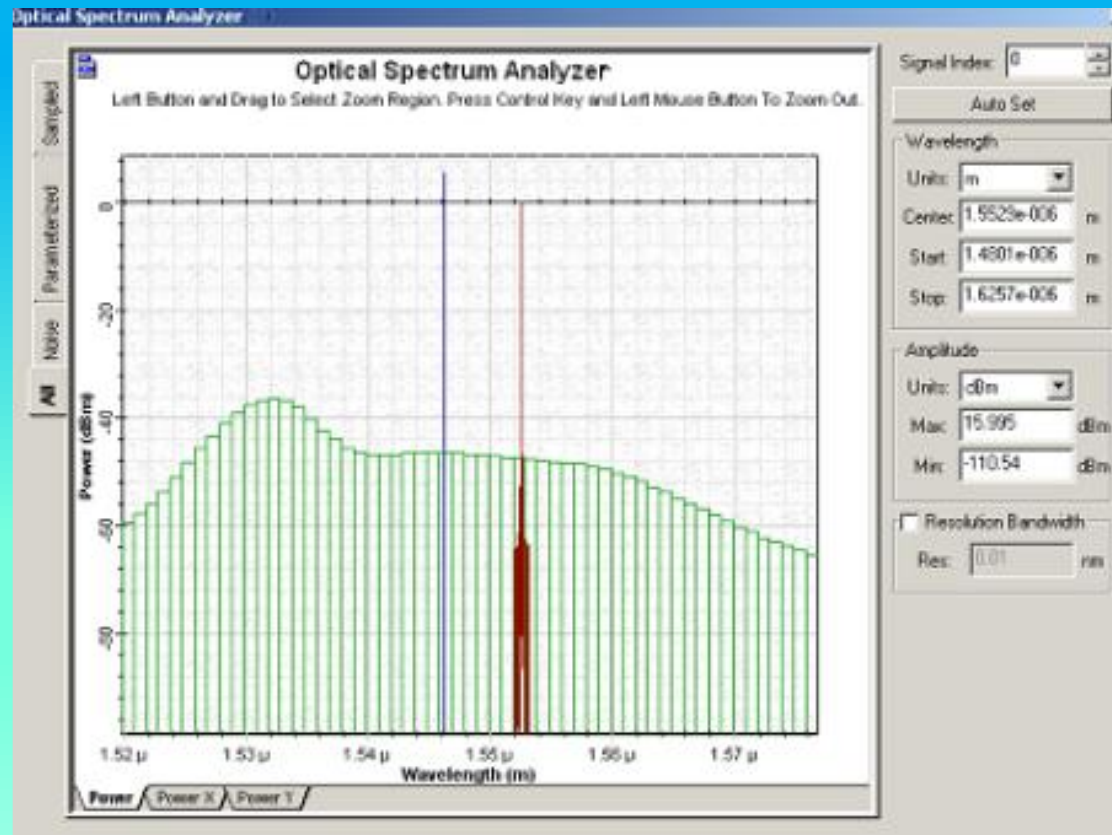


Optical Signal Observed in the OSA

Sampled Signal

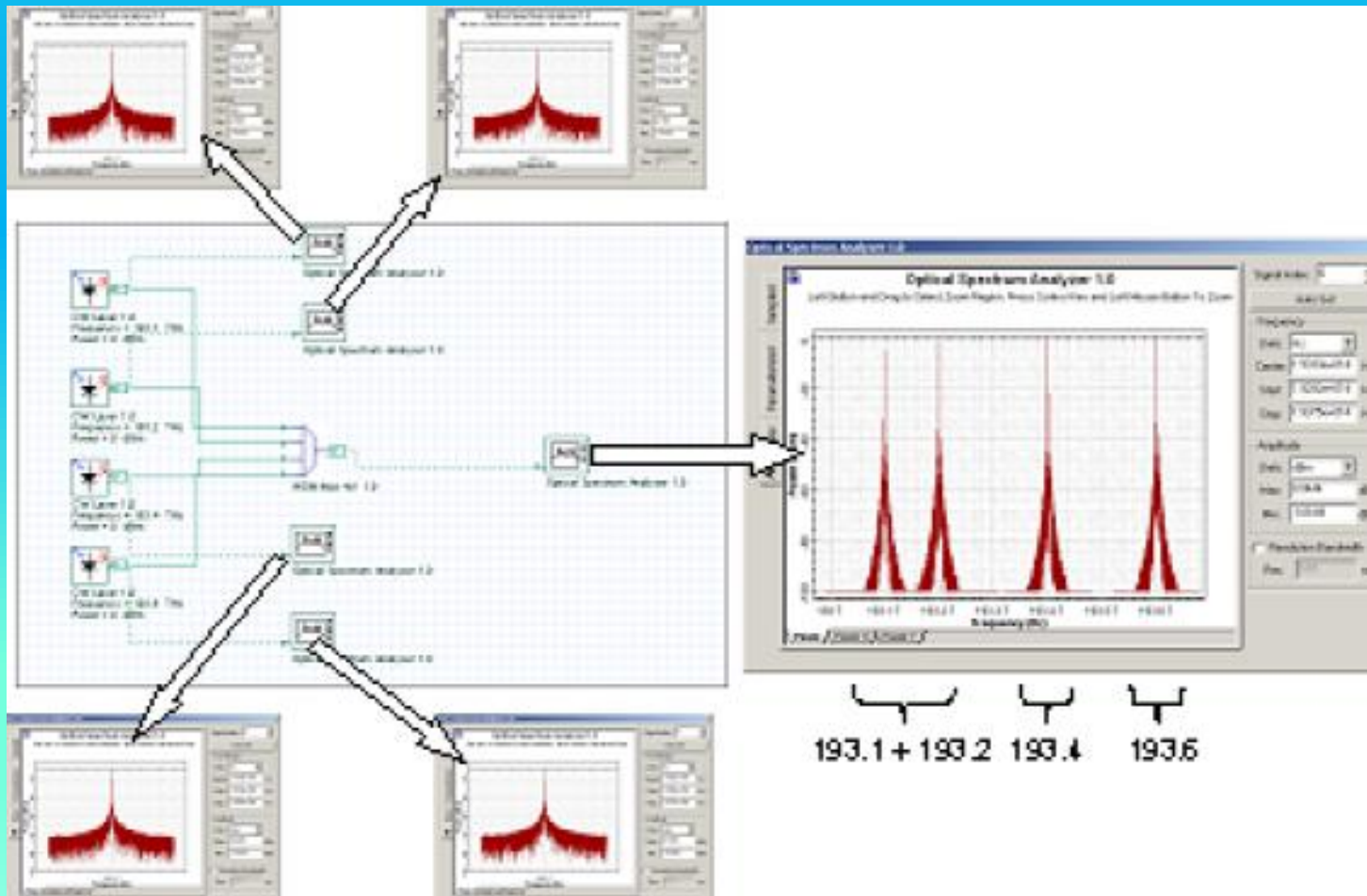
Parametrized

Noise Bins



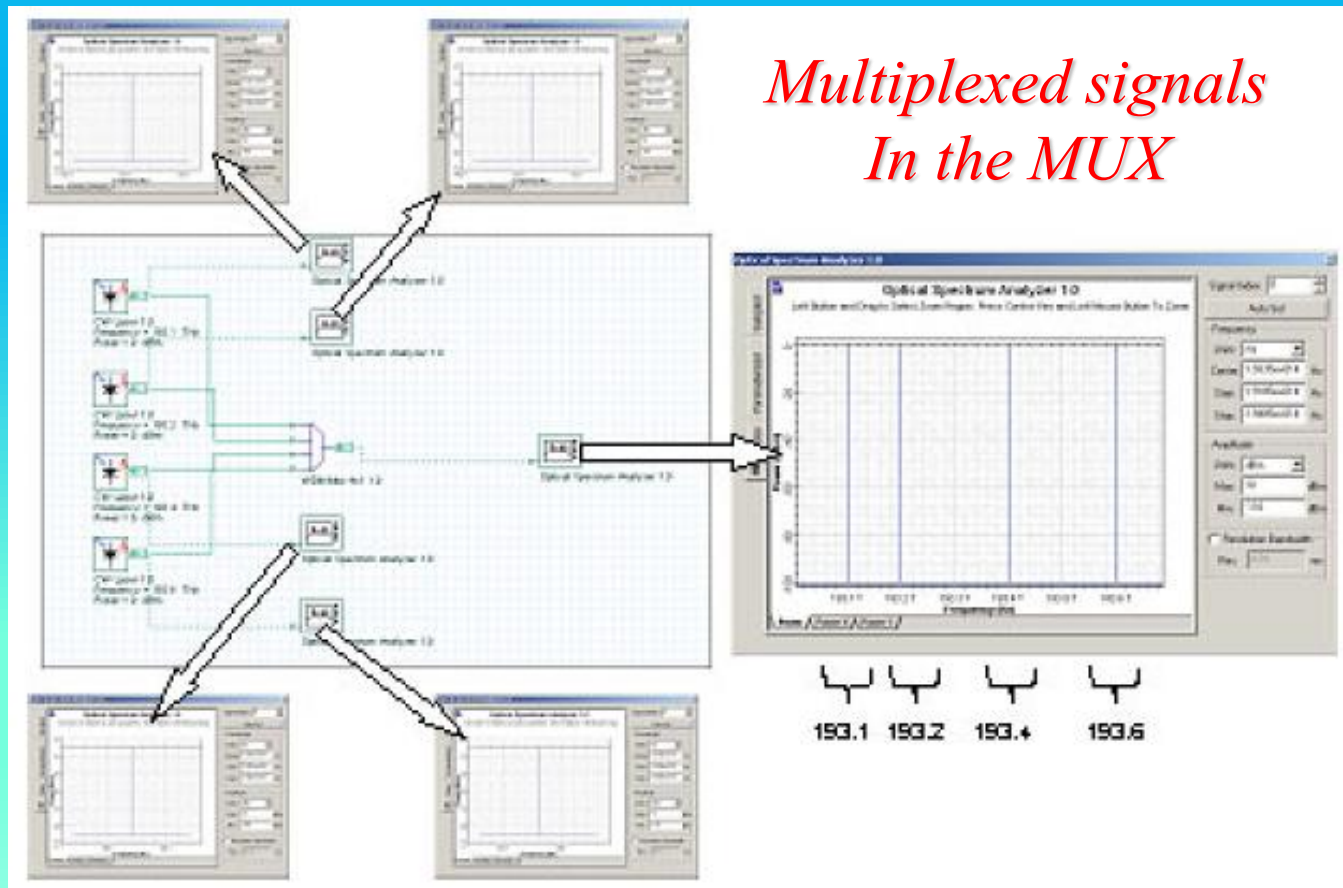


Sampled Signals





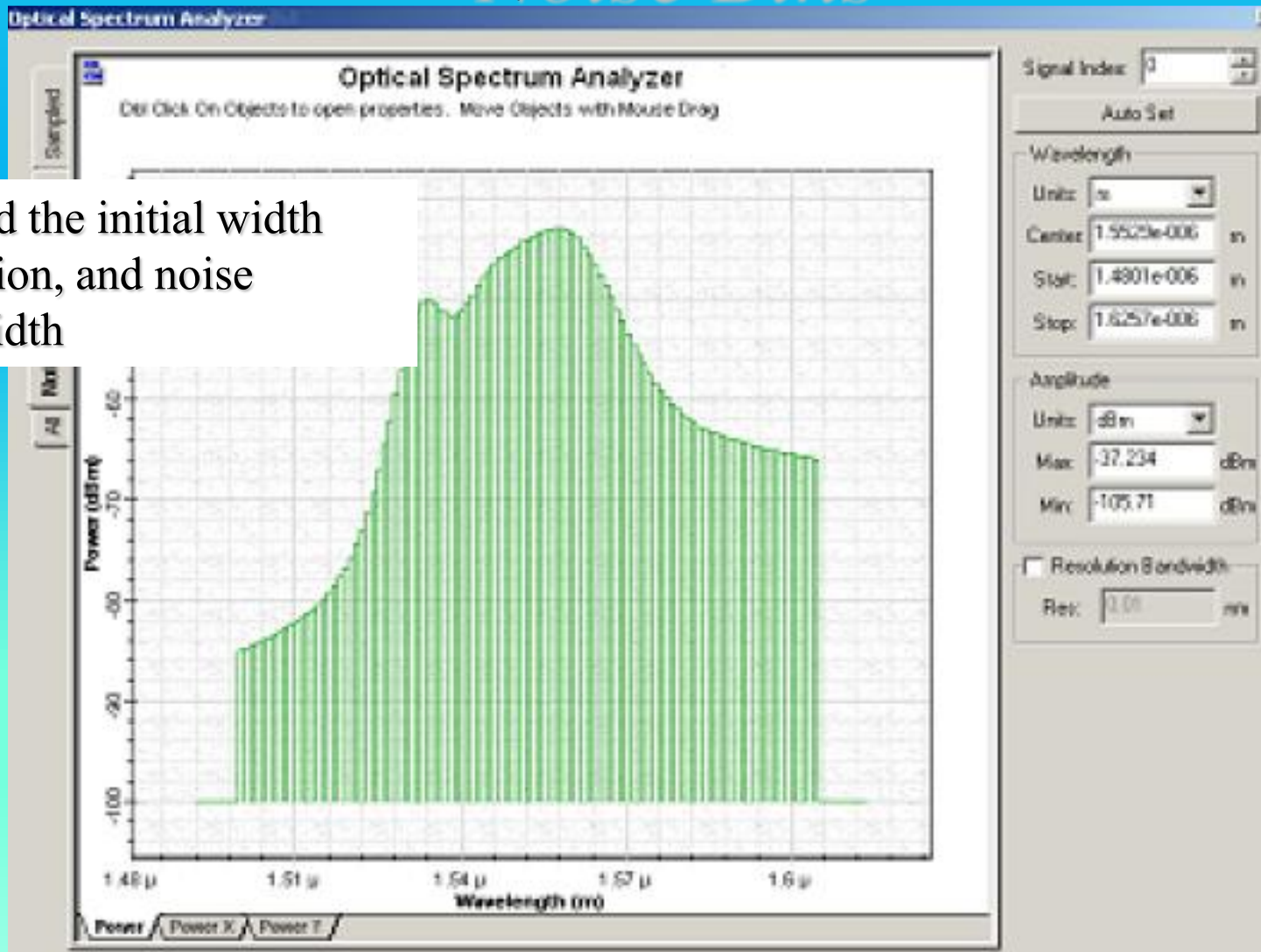
Parametrized Signals





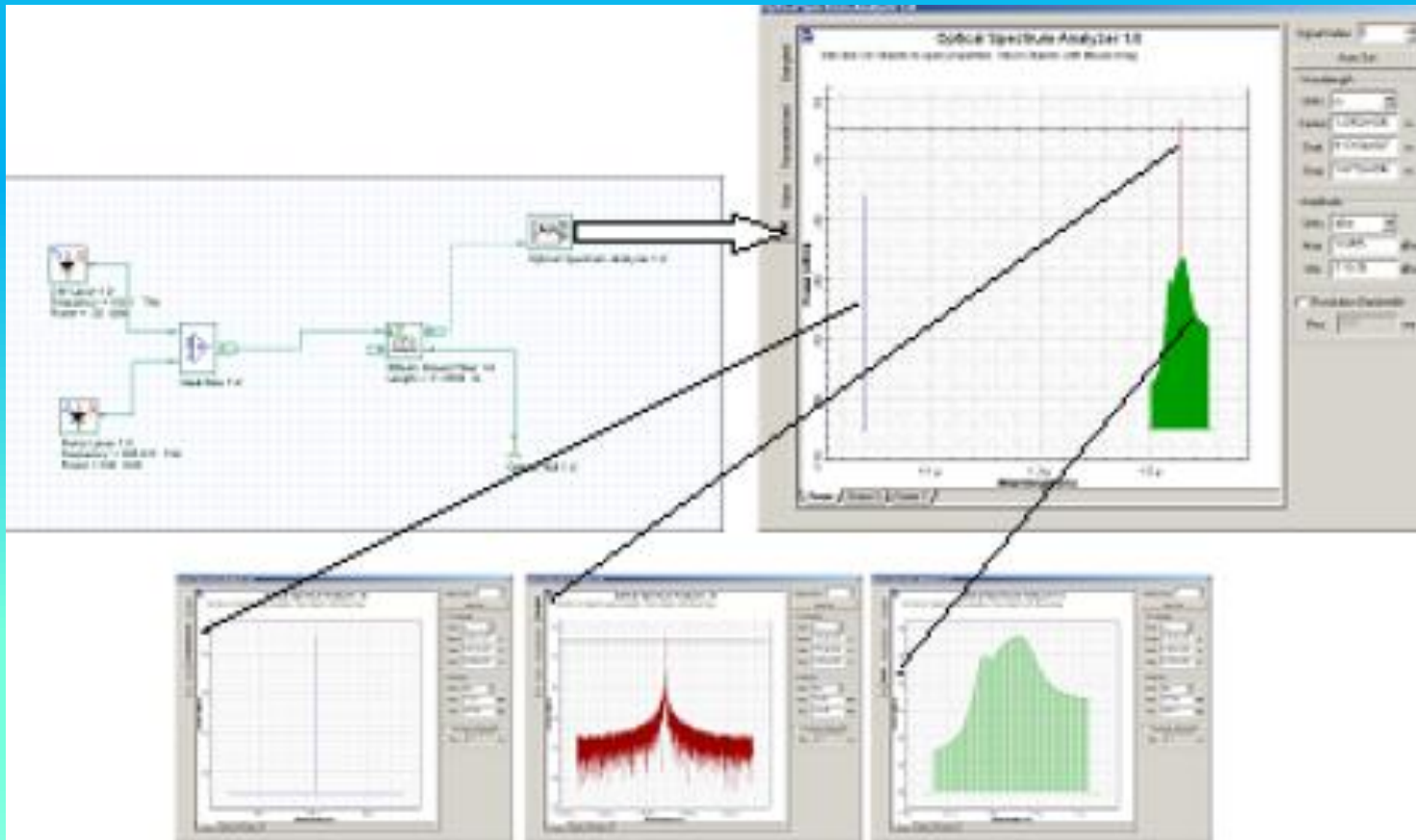
Noise Bins

Defined the initial width
resolution, and noise
bandwidth



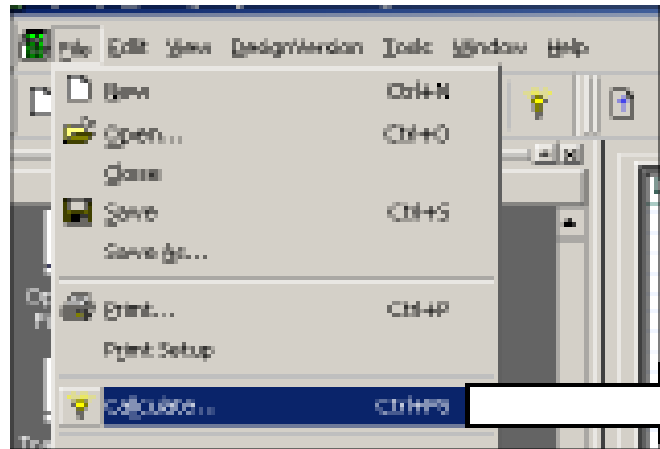


Combined Signals in an EDFA

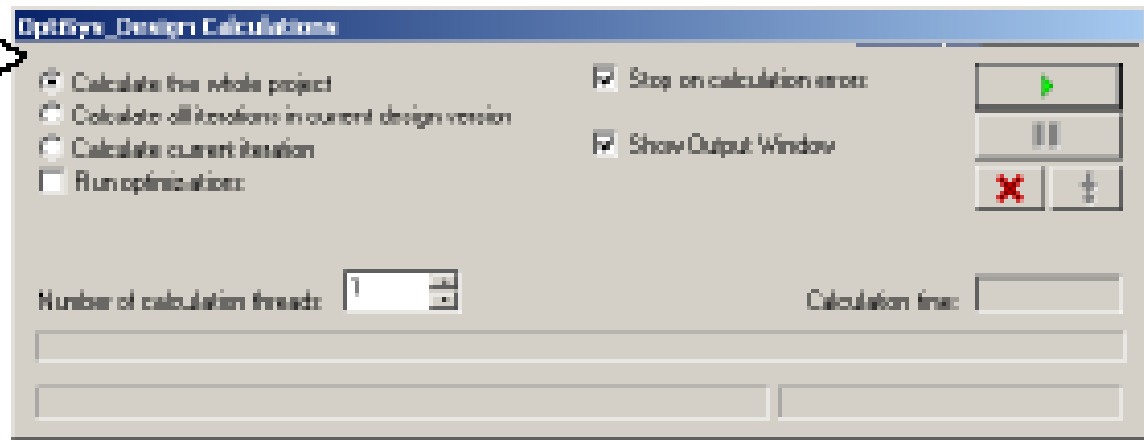




Running a Simulation

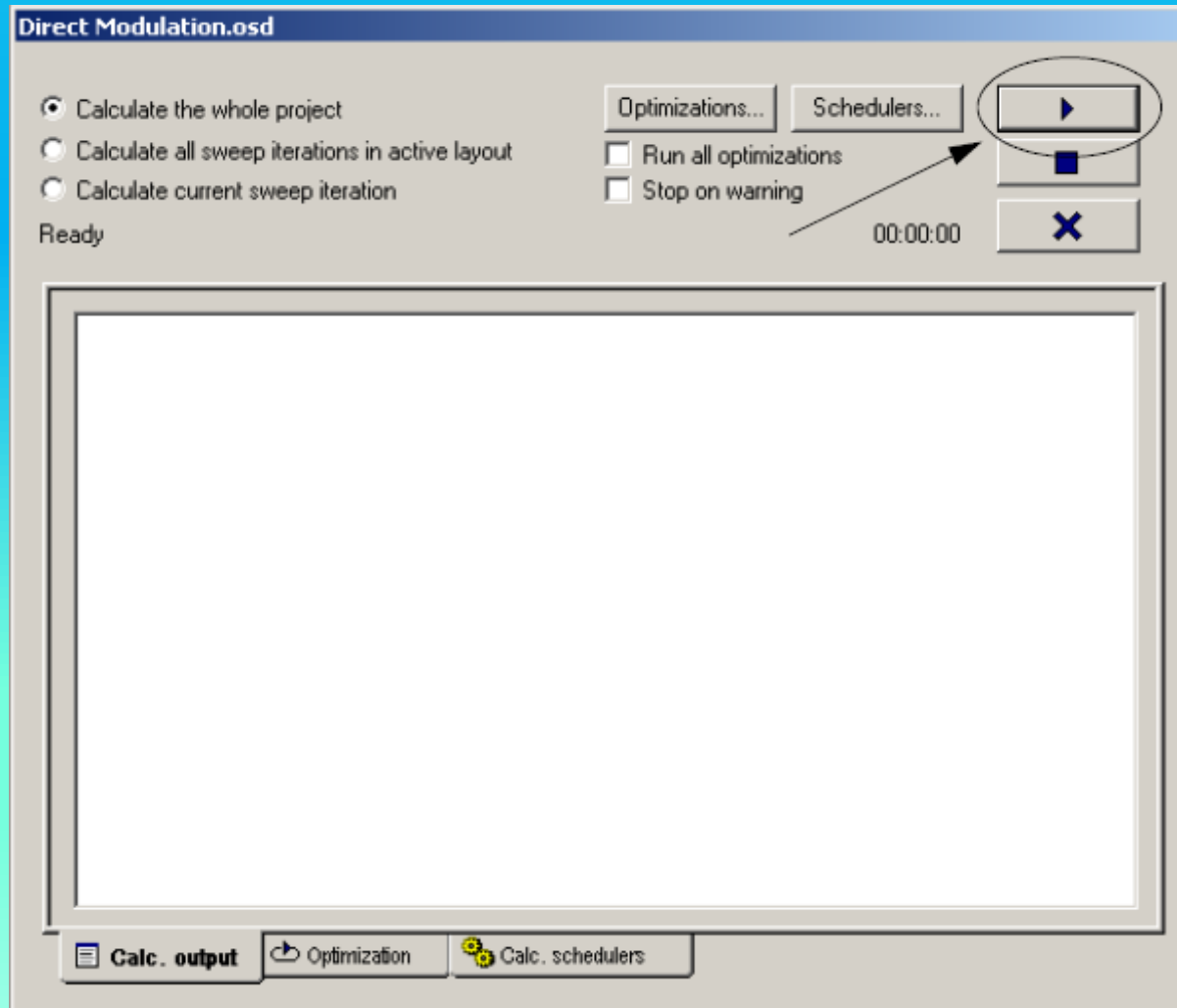


*Report showing the
calculation evolution*





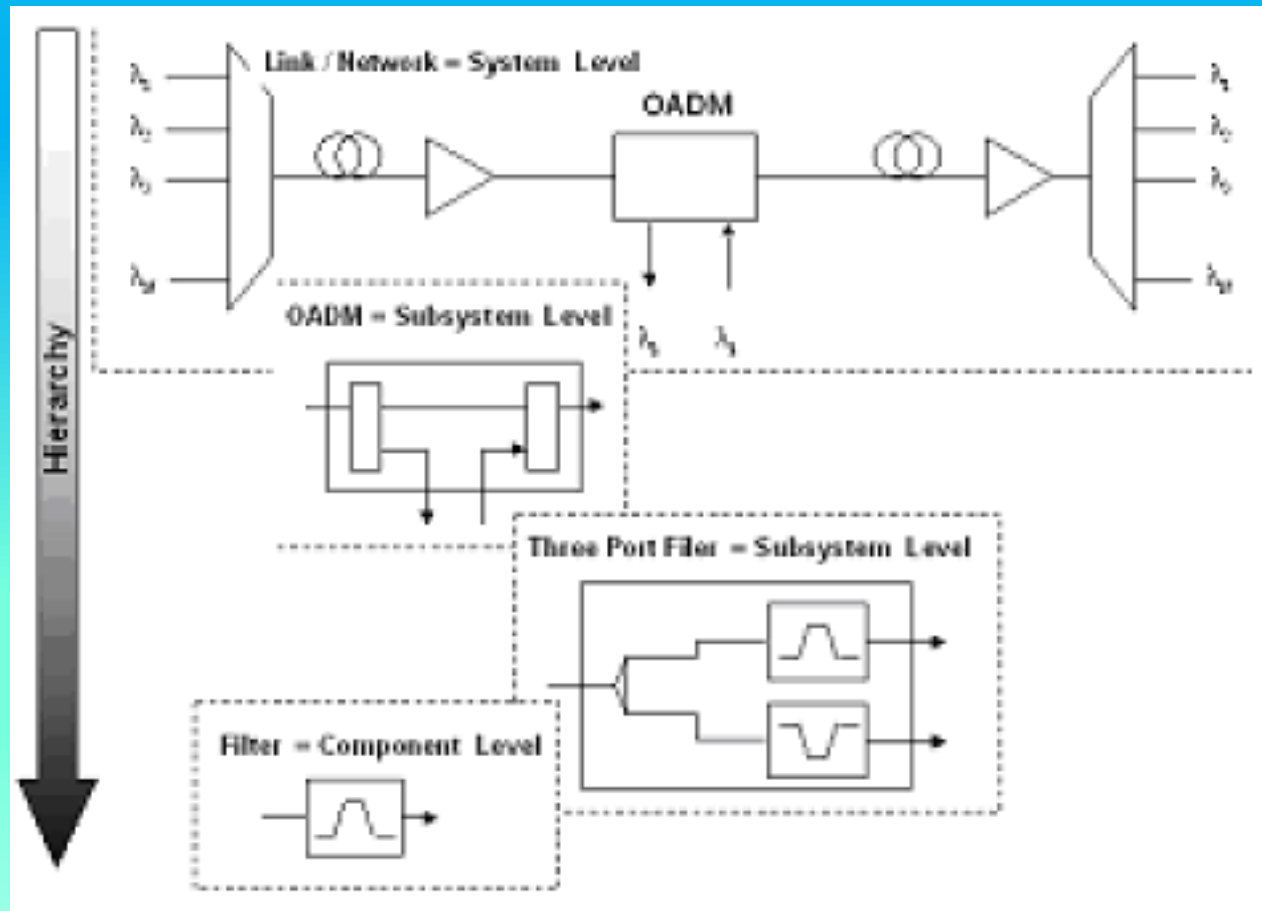
Monitoring the Calculation Evolution





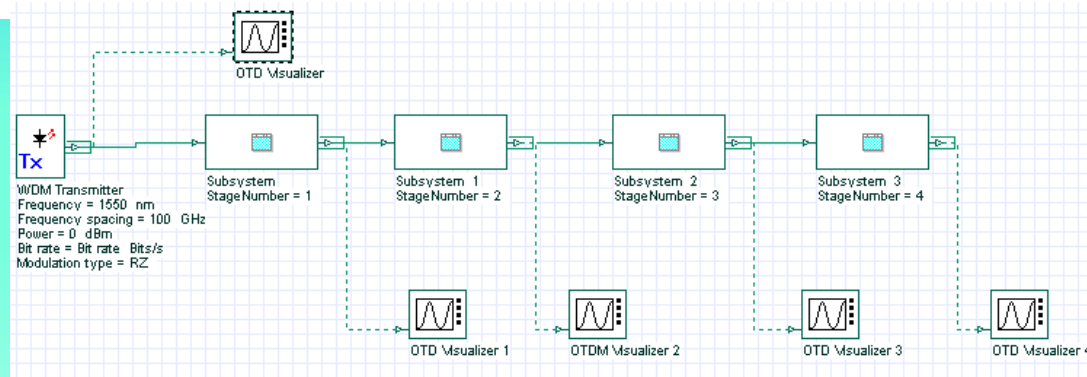
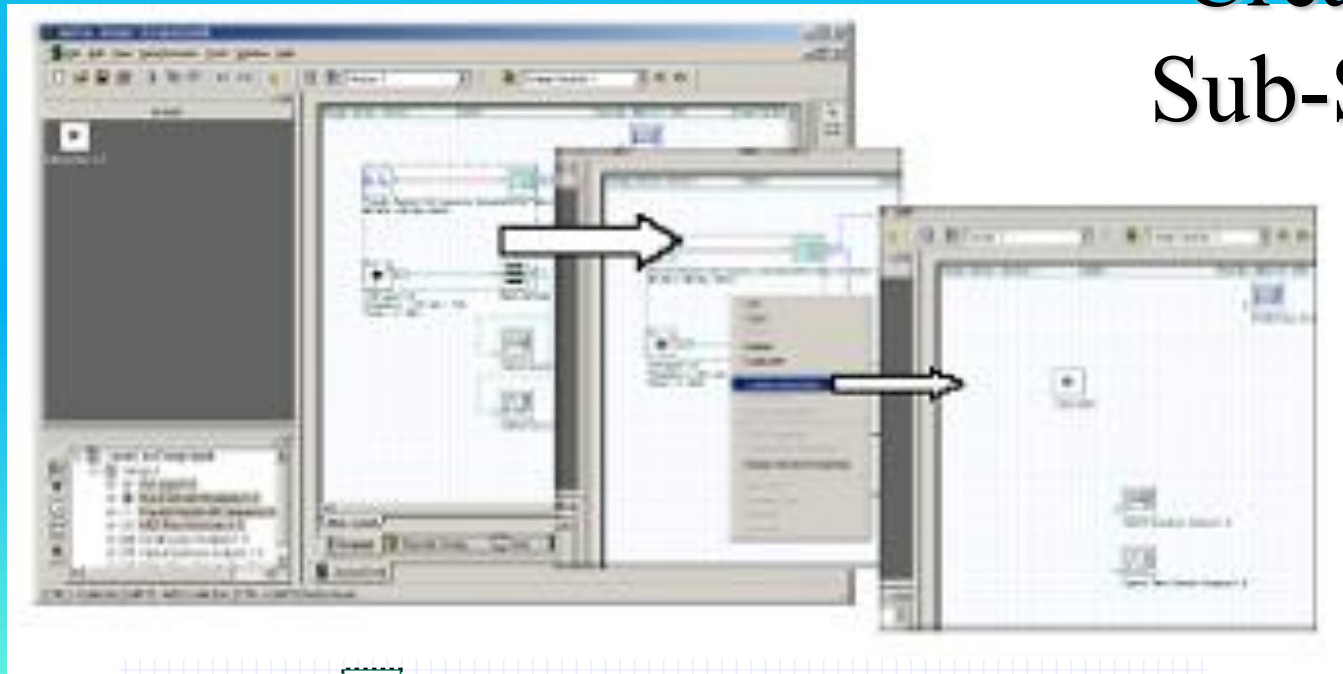
2.1 The Concept of a Sub-System

Enable setting-up your own component



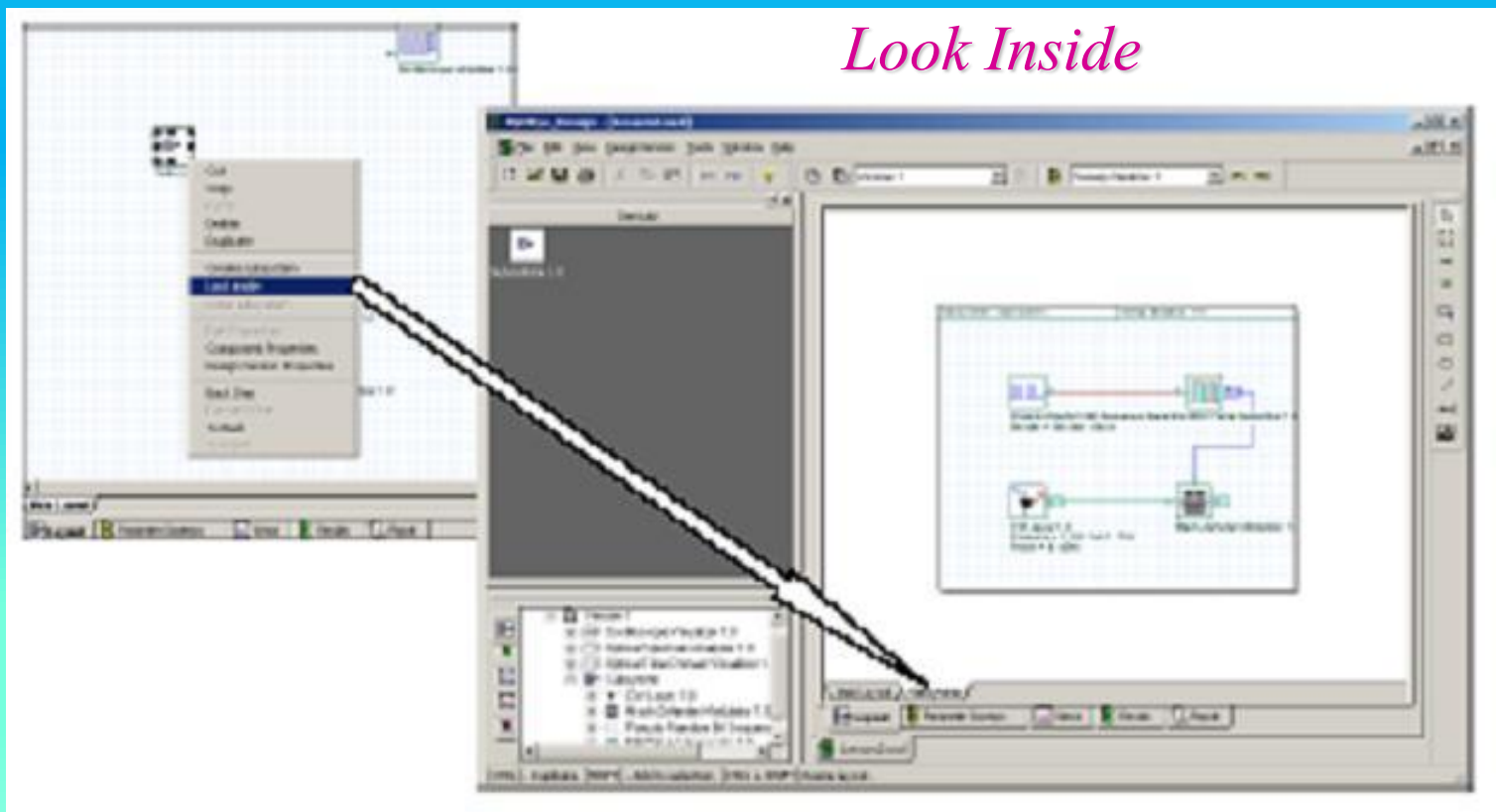


Creating a Sub-System





Go In/Out a Sub-System





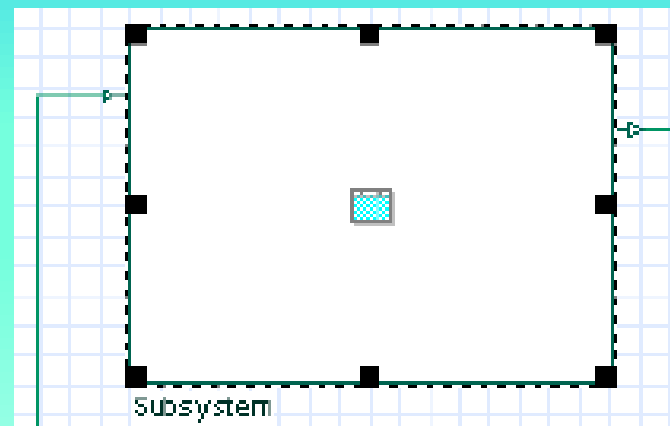
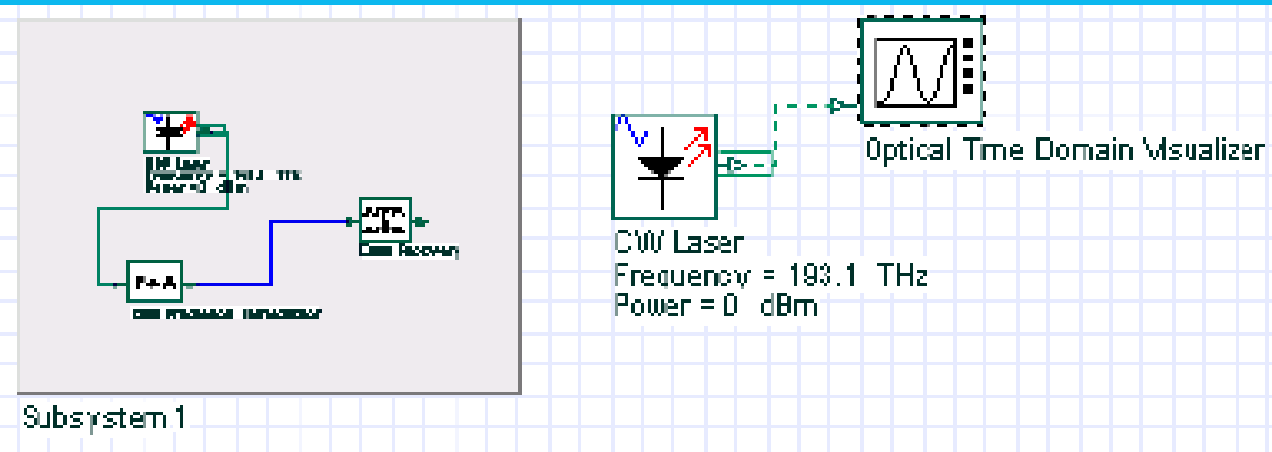
Working with Sub-Systems

Cut
 Copy
 Paste
 Duplicate
 Delete
 Delete All Links
 Select All

Create Subsystem
 Lock Inside
 Close Subsystem

Port Properties...
 Component Properties...
 Layout Properties...
 Bill of Materials...

Back One
 Forward One
 To Back
 To Front



*Grouping
components
and simplify the
layout*

Chances in a Sub-System

Subsystem 1 Properties

Label: Subsystem 1 Cost\$: 0.00

Image

Disp	Name	Value	Units	Mode
☐	Subsystem Representatio	GlassBox Image		Normal
☐	Image Filename	...		Normal
☐	Stretch Image	☐		Normal

Add Parameter

Name: Param1 Add

Type: floating-point Cancel

Category: Image Help

Minimum value: 0

Maximum value: 100

Current value: 0

Units:

Choice list:

OK Cancel Evaluate Script Add Param... Remove Par Edit Param... Load... Save As... Security...

Edit Parameter

Name: Subsystem Representation OK

Type: choice Cancel

Category: Image Help

Minimum value:

Maximum value:

Current value: 0

Units:

Choice list: GlassBox Image, User Image,

OptiSystem 3.0

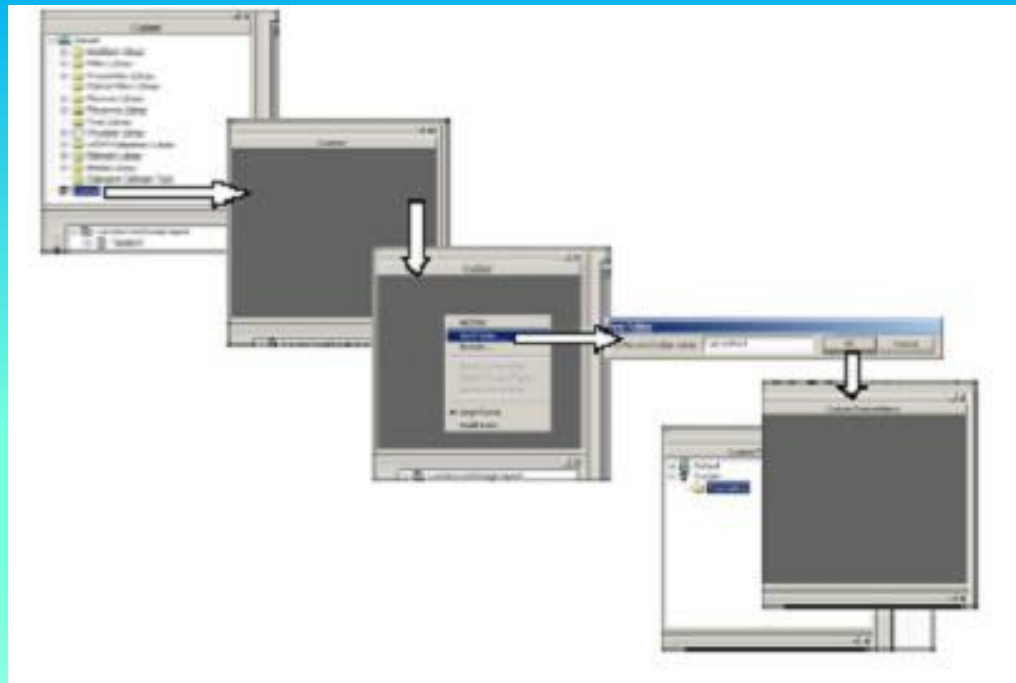
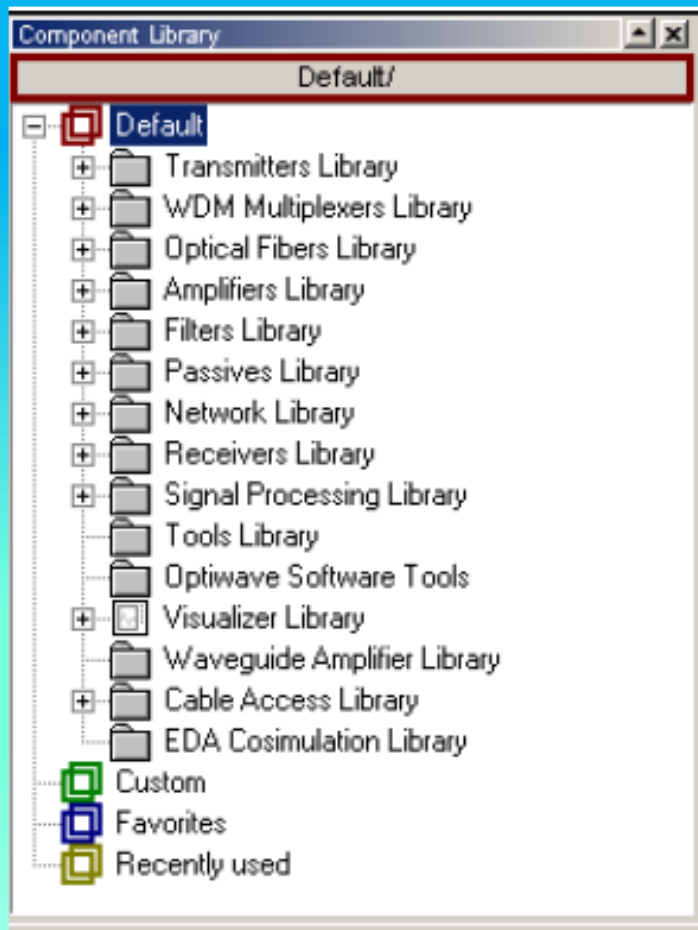
 Remove parameter 'Param1'?

Yes No



3. Component Library

Creating Additional Folders in the Component Library under demand





Global Parameters

Bit rate = 2.5 Gb/s

Sequence Length = 128 bits

Time Window = 5,12e-08

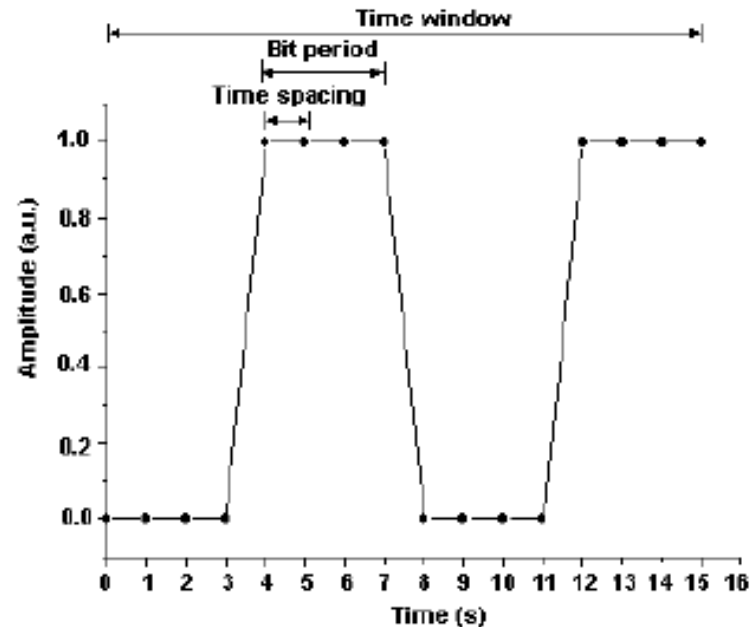
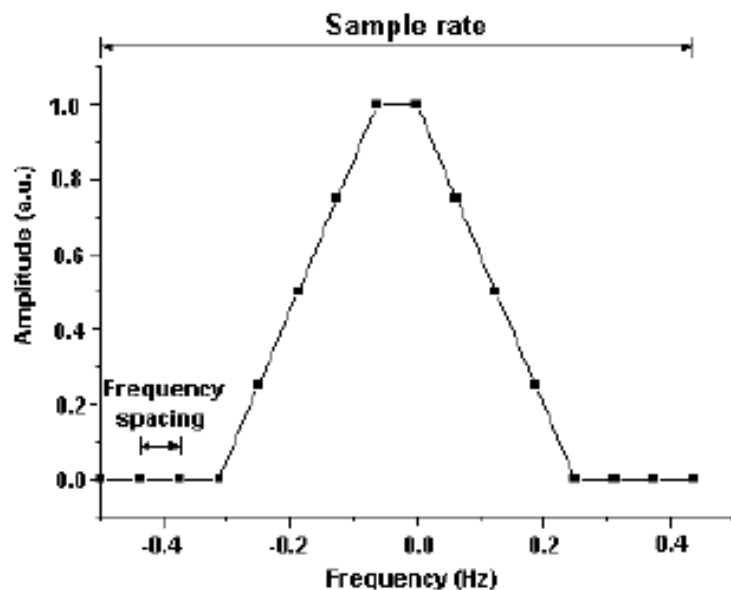
Version 1 Parameters [X]

Label:

Name	Value	Units	Mode
Simulation window	Set bit rate		Normal
Reference bit rate	Set bit rate		Normal
Bit rate	Set time window	Bits/s	Normal
Time window	Set sample rate	s	Normal
Sample rate	1280000000000	Hz	Normal
Sequence length	256	Bits	Normal
Samples per bit	128		Normal
Number of samples	32768		Normal
Iterations	1		Normal



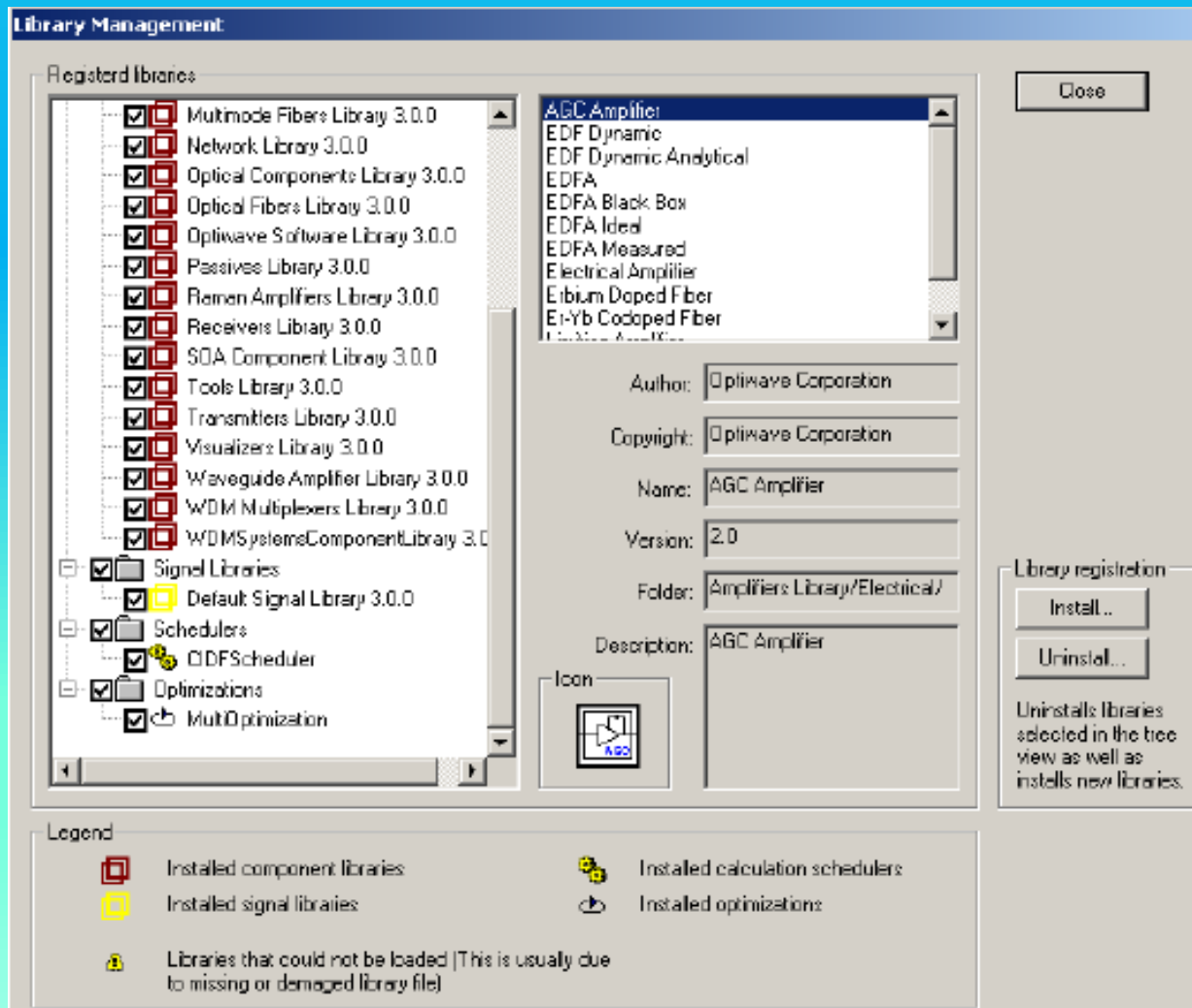
Relating Global Parameters



$\text{Time spacing} = 1 / \text{Sample rate} = \text{Time window} / \text{Number of samples}$
 $\text{Frequency spacing} = 1 / \text{Time window} = \text{Sample rate} / \text{Number of samples}$

$\text{Time window} = \text{Sequence length} * \text{Bit period} = \text{Sequence length} / \text{Bit rate}$
 $\text{Number of samples} = \text{Sequence length} * \text{Samples per bit} = \text{Time window} * \text{Sample rate}$

Managing the Component Library

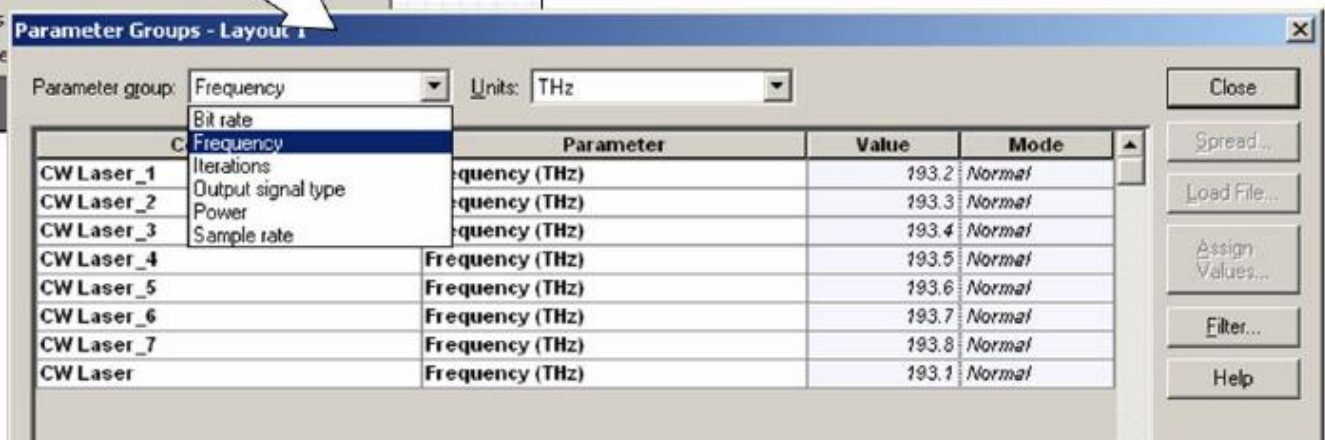


To optimize
memory space:
**Enable or Disable
Libraries**

Group of Parameters



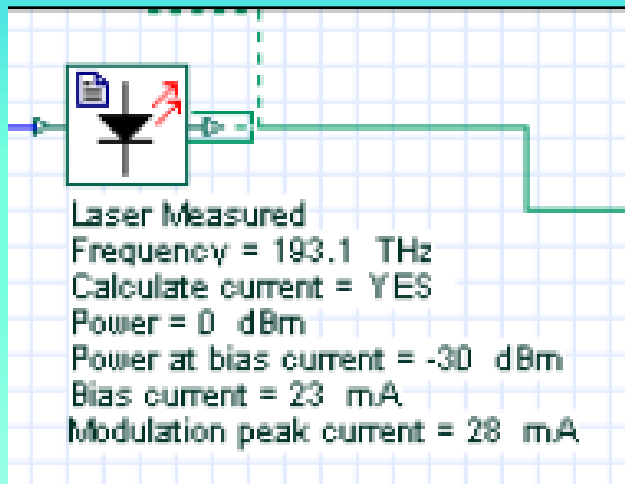
Allows modifying an ensemble of values related to a specific parameter through a unique action





4. Implemented Modeling

- Lasers:
CW (Continuous Wave,
Pulsed,
Measured,
and so on



Laser Measured Properties

Label: Laser Measured Cost\$: 0.00

Main Measure... Physical Initial esti... Simulation Noise Random n...

Disp	Name	Value	Units	Mode
☒	Frequency	193.1	THz	Normal
☐	Calculate current	☒		Normal
☒	Power	0	dBm	Normal
☐	Power at bias current	-30	dBm	Normal
☐	Bias current	23	mA	Normal
☐	Modulation peak current	28	mA	Normal

OK Cancel Evaluate Script Load... Save As... Security... Help



Different Models for a selected Component

Optical

Raman

Raman Amplifier.....	235
Raman Amplifier—Average power model	251
Raman Amplifier—Dynamic model	261

EDFA

EDFA Black Box.....	271
EDF Dynamic — Full model.....	283
EDF Dynamic — Analytical model	289
EDFA.....	295
EDFA Ideal.....	303
EDFA Measured.....	309
Erbium doped fiber.....	315
Er-Yb codoped fiber	353
Er-Yb codoped waveguide amplifier	363

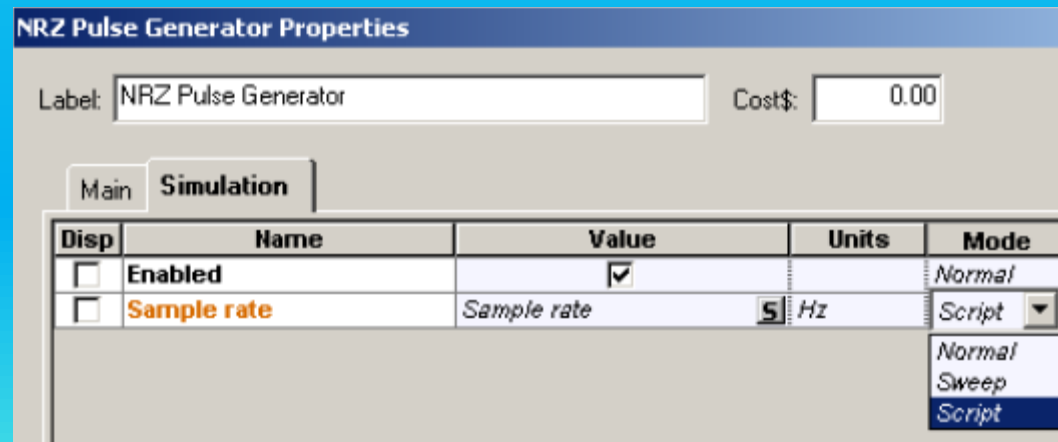
SOA

Semiconductor Optical Amplifier	381
---------------------------------------	-----

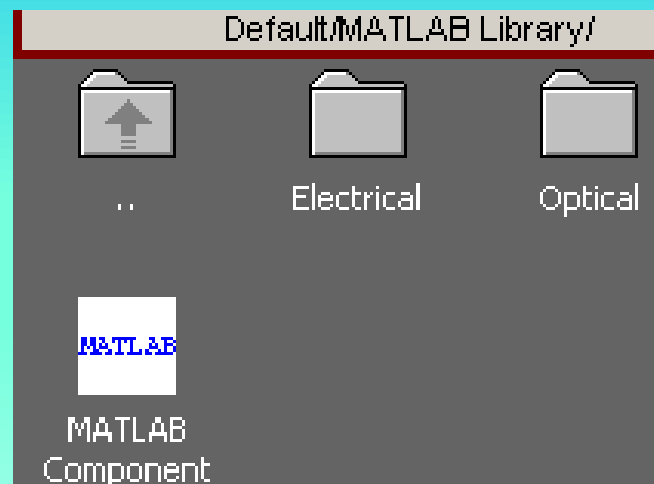


Including your Own Model

Script



Co-Simulation





Using the Script

```
'Get Layout Manager.  
Dim Lm  
Set Lm = Document.GetLayoutMgr  
  
'SCRIPT for Version 1  
  
'Get Current Layout.  
Dim Layout1  
Set Layout1 = Lm.GetCurrentLayout  
Layout1.Name = "Version 1"  
  'Set Total Sweep Iterations  
Layout1.SetTotalSweepIterations(10)  
  'Set Current Sweep Iteration  
Layout1.SetCurrentSweepIteration(10)  
'Get Current Canvas.  
Dim Canvas1  
Set Canvas1 = Layout1.GetCurrentCanvas  
'SCRIPT for Layout global parameters.  
Layout1.SetParameterMode "Simulation window", 0  
Layout1.SetParameterValue "Simulation window", "Set bit rate"  
Layout1.SetParameterMode "Reference bit rate", 0  
Layout1.SetParameterValue "Reference bit rate", TRUE  
Layout1.SetParameterMode "Bit rate", 0  
Layout1.SetParameterValue "Bit rate", 1e+010  
Layout1.SetParameterMode "Time window", 0  
Layout1.SetParameterValue "Time window", 2.56e-008  
Layout1.SetParameterMode "Sample rate", 0  
Layout1.SetParameterValue "Sample rate", 1.28e+012  
Layout1.SetParameterMode "Sequence length", 0  
Layout1.SetParameterValue "Sequence length", 256  
Layout1.SetParameterMode "Samples per bit", 0
```

Allows modification:

- global parameters
- layout
- components' parameters



Exercise 4

4a. Find the lasers in the Component Library

4b. Characterize 3 diferentes lasers available in the Laser's Component Library

5. Scanning Parameters Automatically (*Sweep*)



Layout Tools Report Script Add-Ins Window Help

- Add Layout
- Duplicate Layout
- Delete Layout
- Set Current Sweep Iteration...
- Set Total Sweep Iterations... Ctrl + Home
- Previous Sweep Iteration Ctrl + Page Up
- Next Sweep Iteration Ctrl + Page Down
- Parameter sweeps...
- Parameter Groups...
- Layout Size...
- Parameters...
- Properties...
- Bill of Materials...

Total Parameter Sweep Iterations

Total iterations: OK

Current iteration: Cancel

CW Laser Properties

Label: CW Laser Cost\$: 0.00

Main Polarization Simulation Noise Random numbers

Disp	Name	Value	Units	Mode
☒	Frequency	1576.315789474	nm	Normal
☒	Power	-13.68421052632	dBm	Sweep
☐	Linewidth	10	MHz	Normal
☐	Initial phase	0	deg	Normal

OK Cancel Evaluate Script

Choosing the Interval to Sweep Parameters

Parameter Sweeps

	CW Laser
Sweeps	Power [dBm]
1	0
2	-1.052631578947
3	-2.105263157895
4	-3.157894736842
5	-4.210526315789
6	-5.263157894737
7	-6.315789473684
8	-7.368421052632
9	-8.421052631579
10	-9.473684210526
11	-10.52631578947
12	-11.57894736842
13	-12.63157894737
14	-13.68421052632
15	-14.73684210526
16	-15.78947368421
17	-16.84210526316
18	-17.89473684211
19	-18.94736842105
20	-20

OK
Cancel
Help
Assign Values...

Spread Tools
Linear...
Exp...
Log...
Formula...
Gaussian...
LogNormal...
Uniform...
Discrete...

Parameter Sweep Script Editor

Input:
☐ Use sweeps iteration number as input
☒ Get input from range

Minimum:
Maximum:

OK
Cancel
Help

Type "Input" for the input parameter name. E.g. "Input + 2".

Script:

Gaussian sweep iteration

Random seed index:

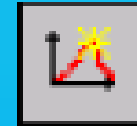
Format for Std. Dev: ☐ absolute ☒ %

Nominal Value:
Std. Dev.

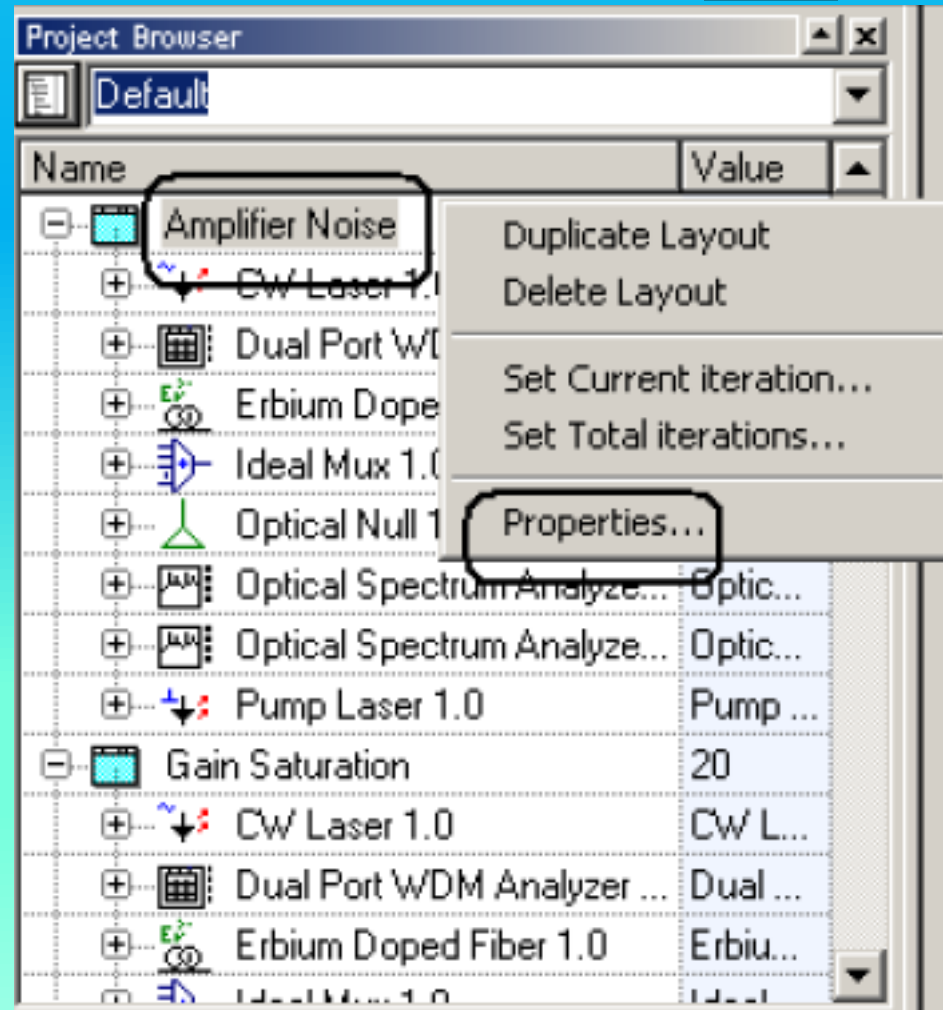
OK
Cancel



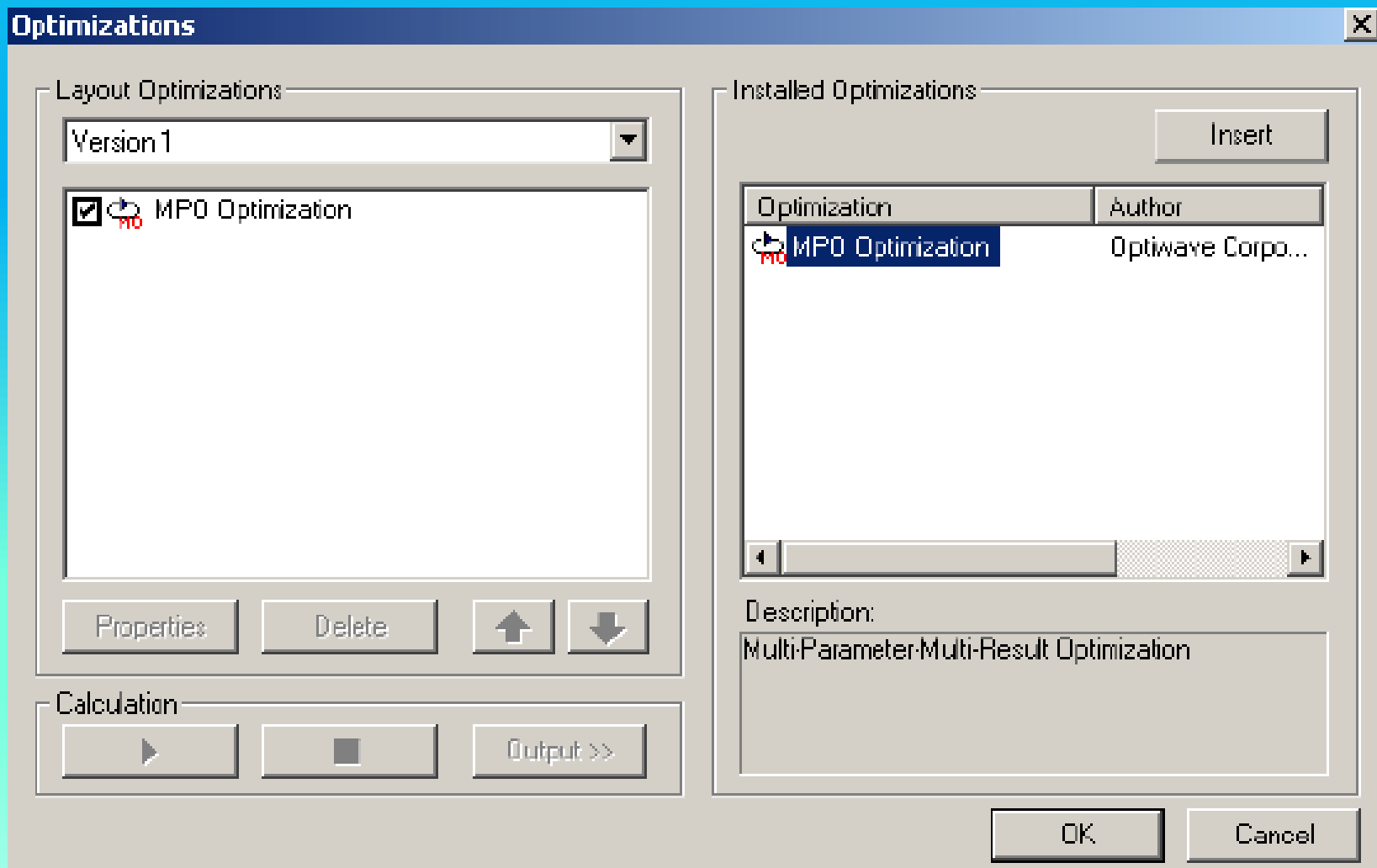
6. Optimization Routines



- *One Parameter*
- *Multi-parameters*
- . *Parameters*
- . *Results*



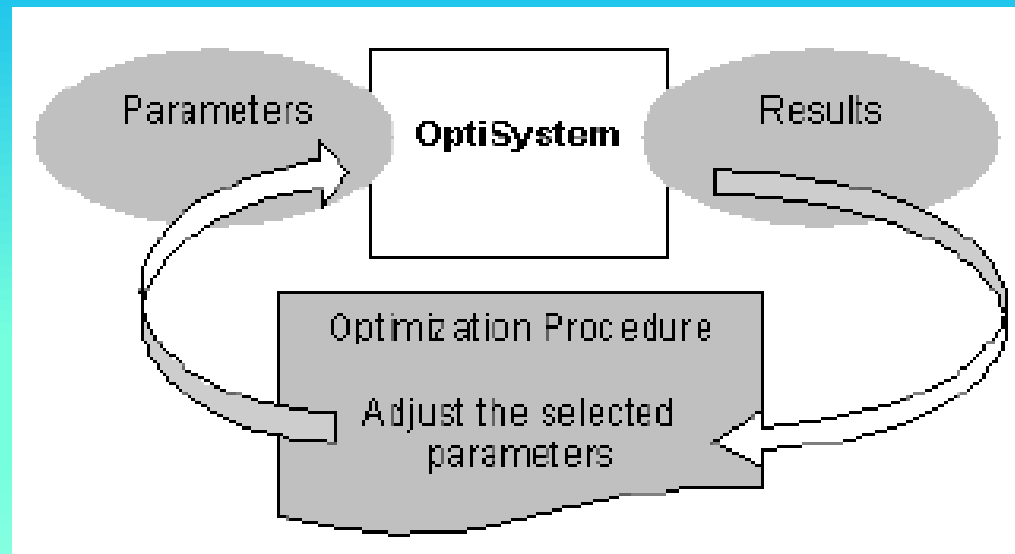
How to Perform the Optimization





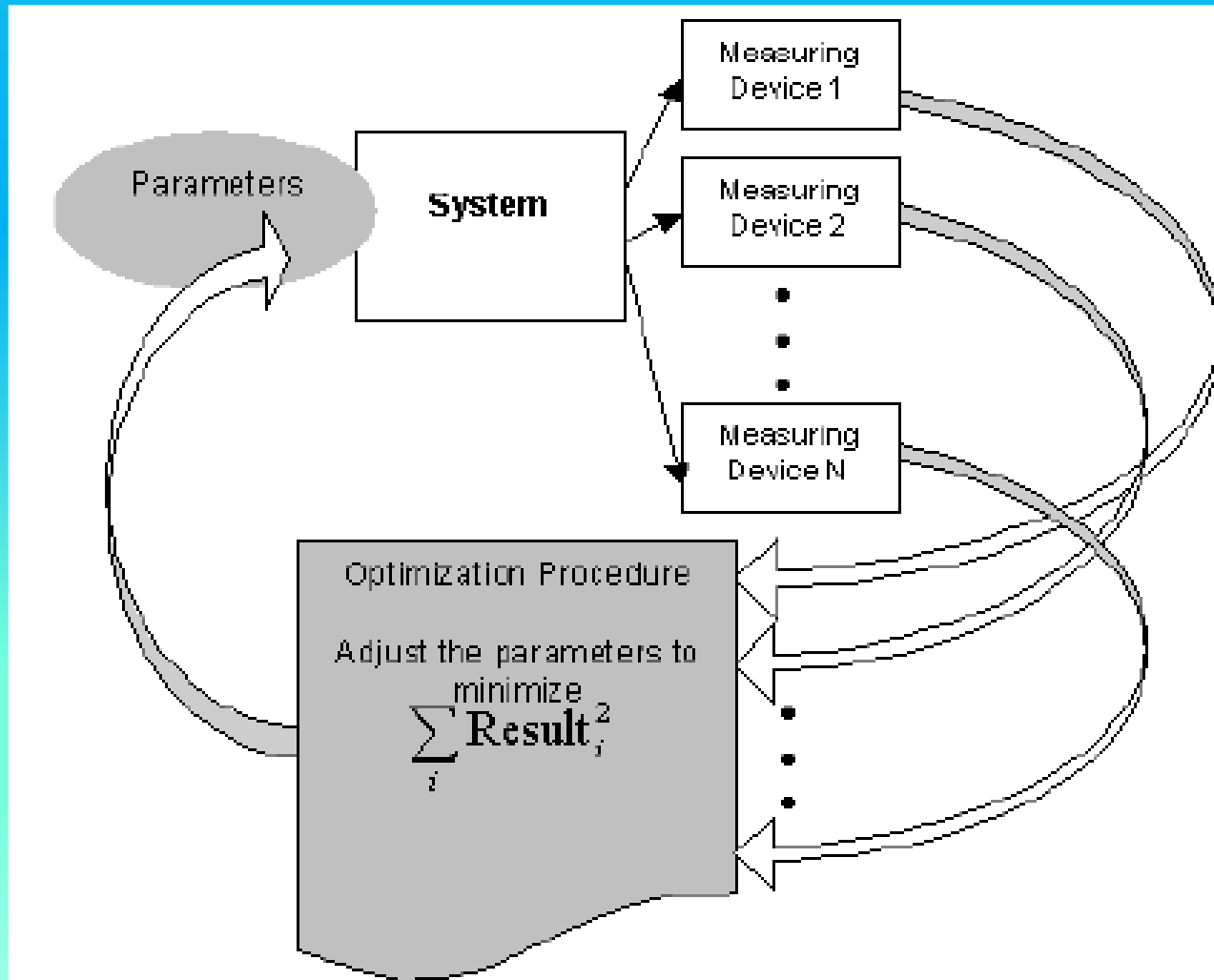
Multiparameter Optimization

Minimize
Maximize
Search based on a set value



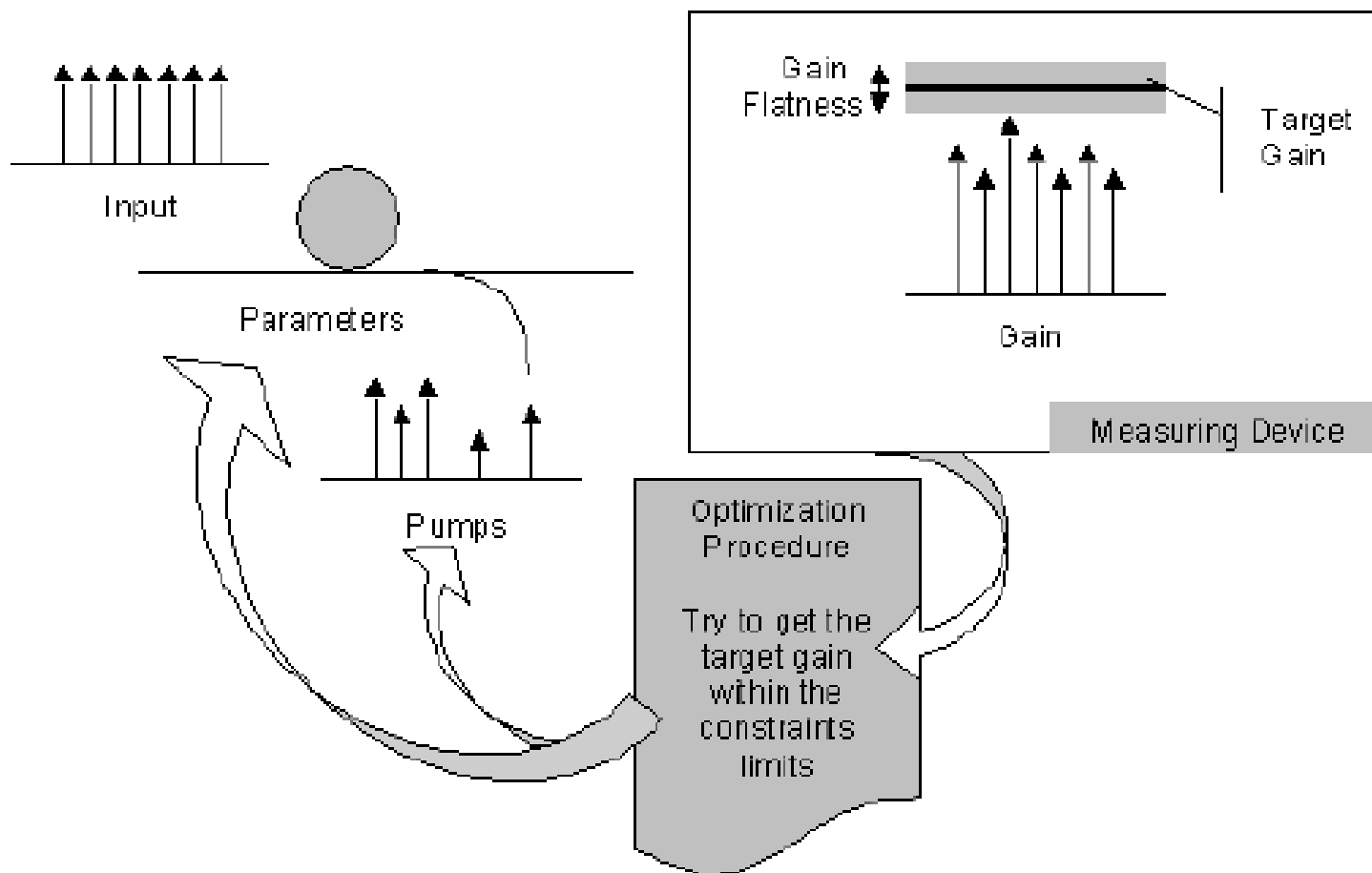


Nonlinear Fitting





Optimization to Obtain Flat Gain



Starting Optimization

Setup

Optimization Name:

Main | Parameters | Results | Advanced

Optimization

Optimization Type:

Goal Attainment:

Maximum number of passes:

Result Tolerance for Exact Goal Attainment (%):

☐ Constrained

General Info

Author:

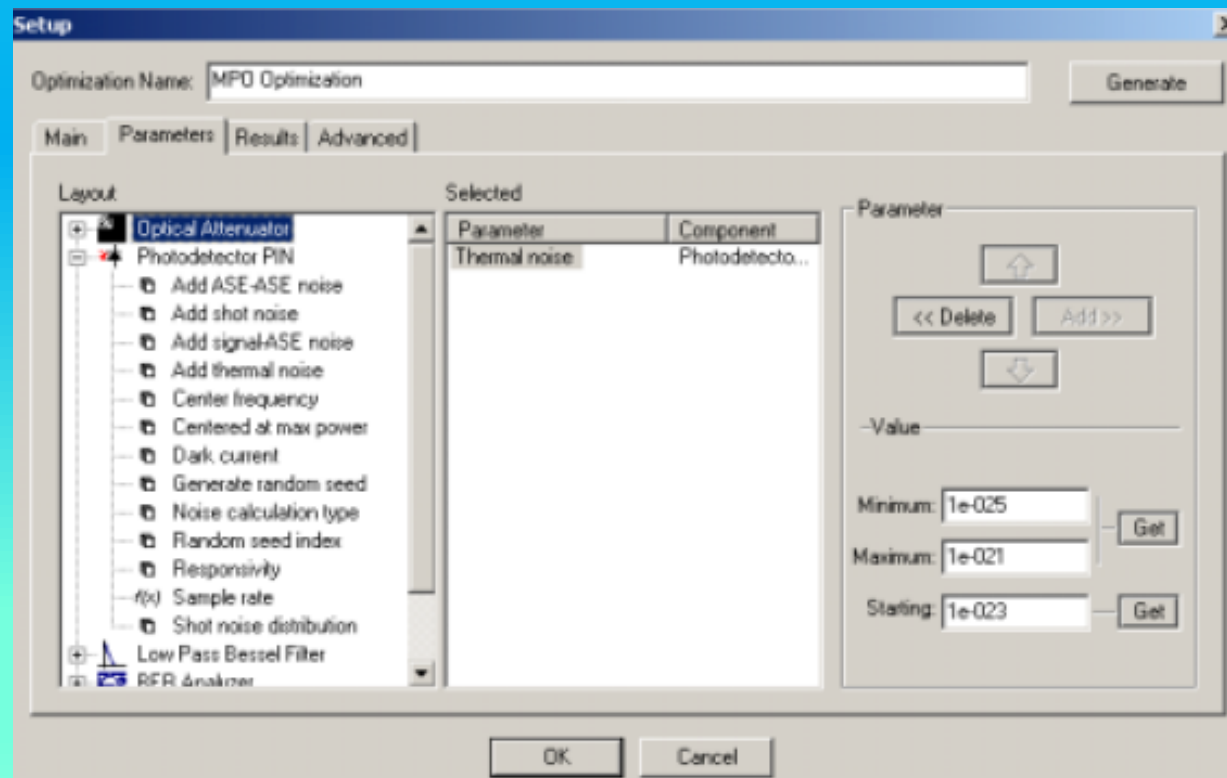
Version:

Copyright:

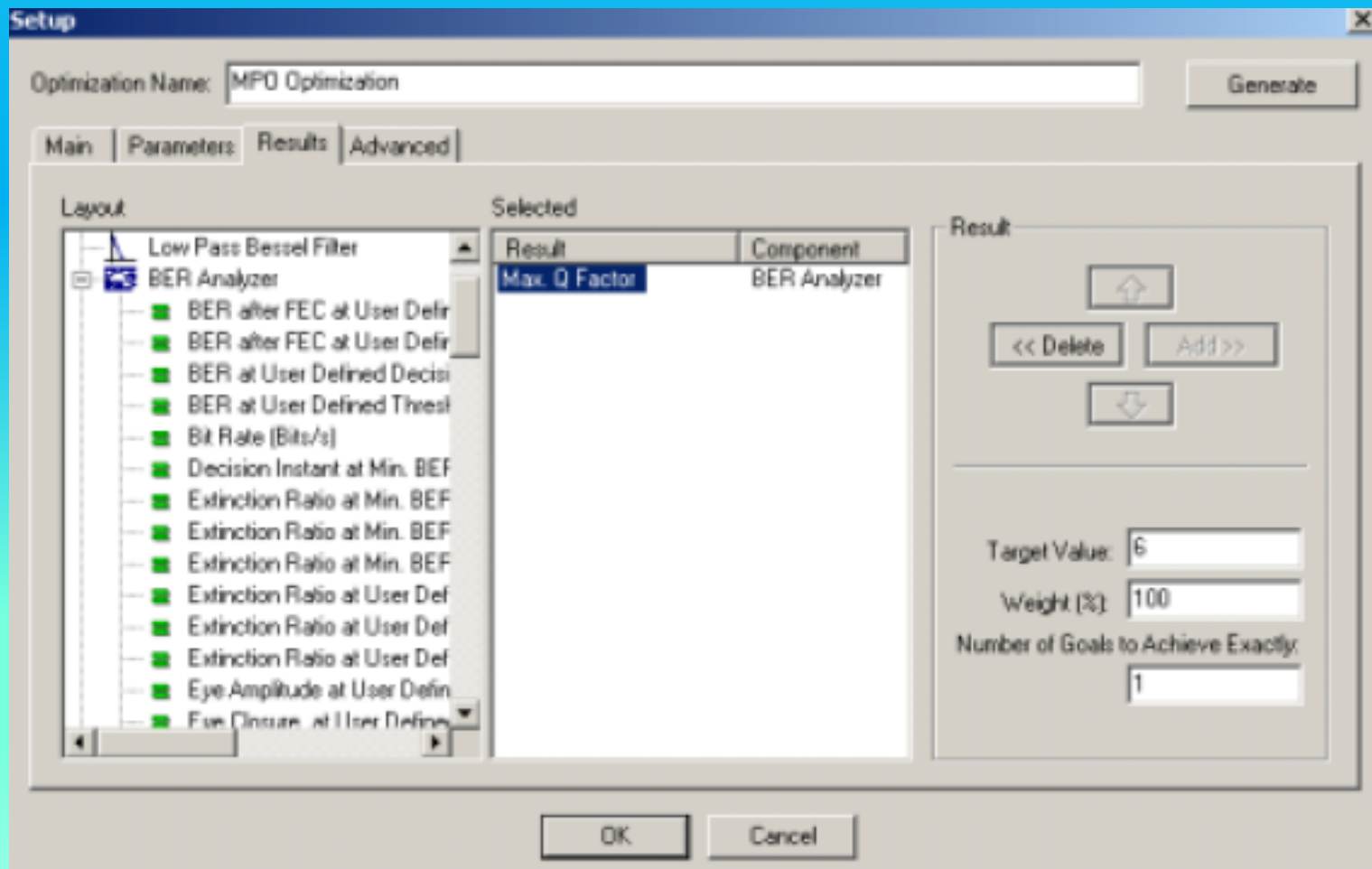
Description:



Choosing Parameters

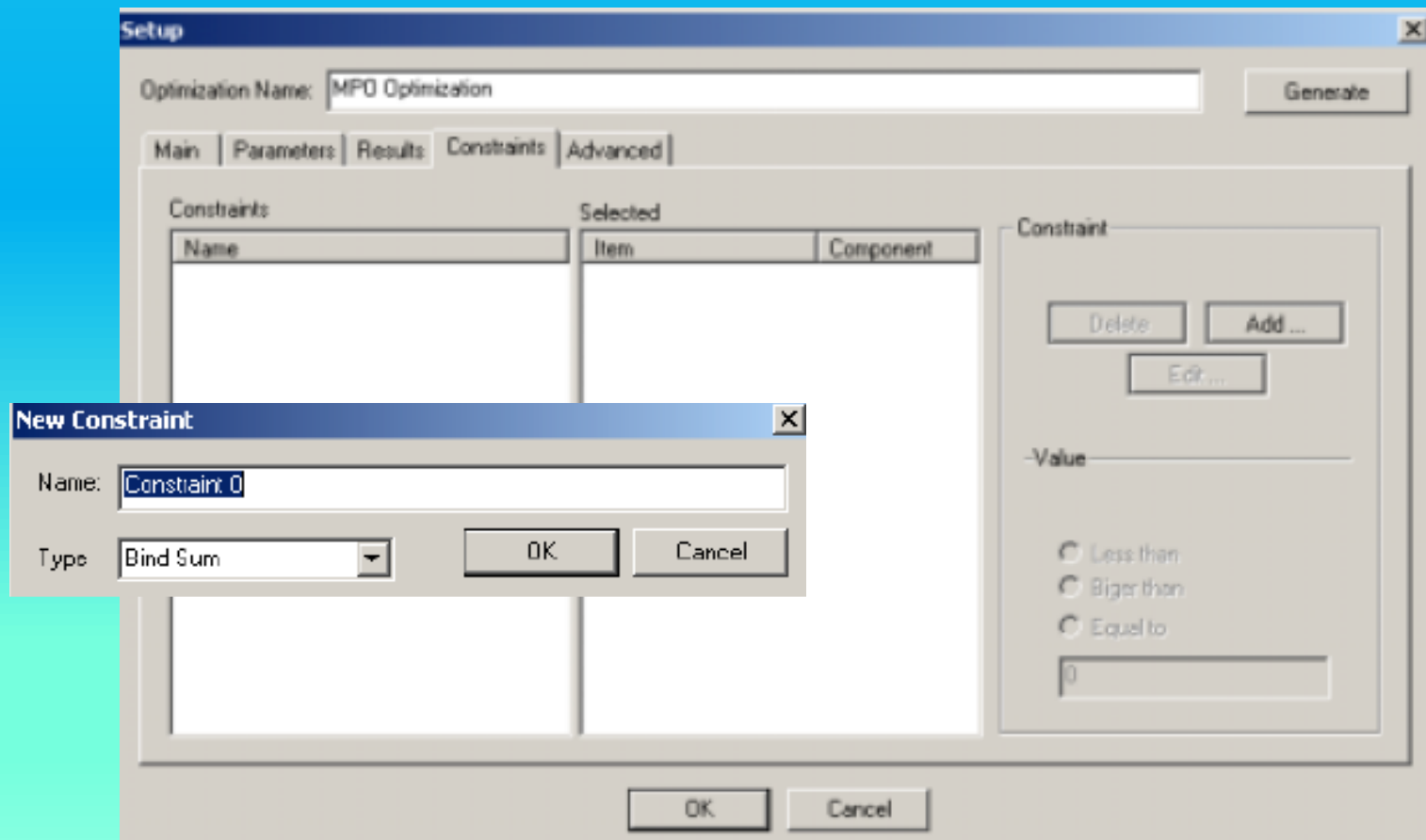


Checking the Results





Adding Constrains





Parameters and Selected Results

Select Parameters

Selected Parameters		Constraint	
Parameter	Component	Parameter	Component
Calculate parameters	Laser Measured	Σ Calculate 3D graph	BER Analyzer

Select Results

Layout Results

- Bit Rate (Bits/s)
- Decision Instant at Min. BER (bit p
- Extinction Ratio at Min. BER
- Extinction Ratio at Min. BER (%)
- Extinction Ratio at Min. BER (dB)
- Extinction Ratio at User Defined D
- Extinction Ratio at User Defined D
- Extinction Ratio at User Defined D
- Eye Amplitude at User Defined De
- Eye Closure at User Defined Deci
- Eye Height at User Defined Decisi
- Eye Opening Factor at User Defin
- Eye Opening Factor at User Defin
- Eye Opening Factor at User Defin
- log of BER after FEC at User Defir
- log of BER after FEC at User Defir
- log of BER at User Defined Decisi

Selected Result

Result	Component
Threshold at Min. BER...	BER Analyzer

Σ

Add

Delete

Done



7. Vizualizers: Electrical and Optical

- Optical

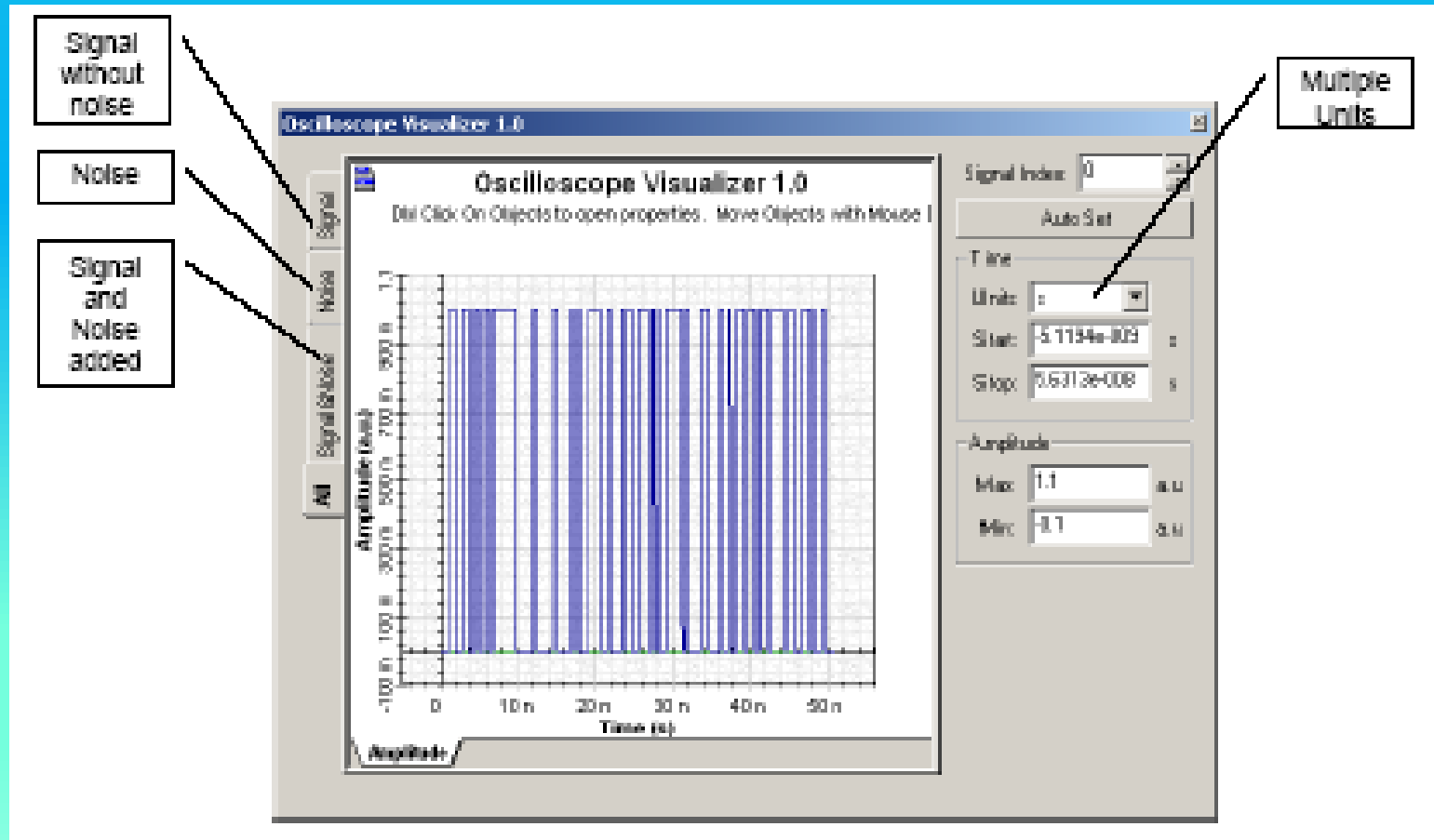
- OSA;
- Time Domain Visualizer;
- Optical Power Meter;
- Dual Port WDM Analyzer;

- Electrical

- Oscilloscope;
- Spectrum Analyzer
- (Electrical) Power Meter;
- Eye Diagram Analyzer;

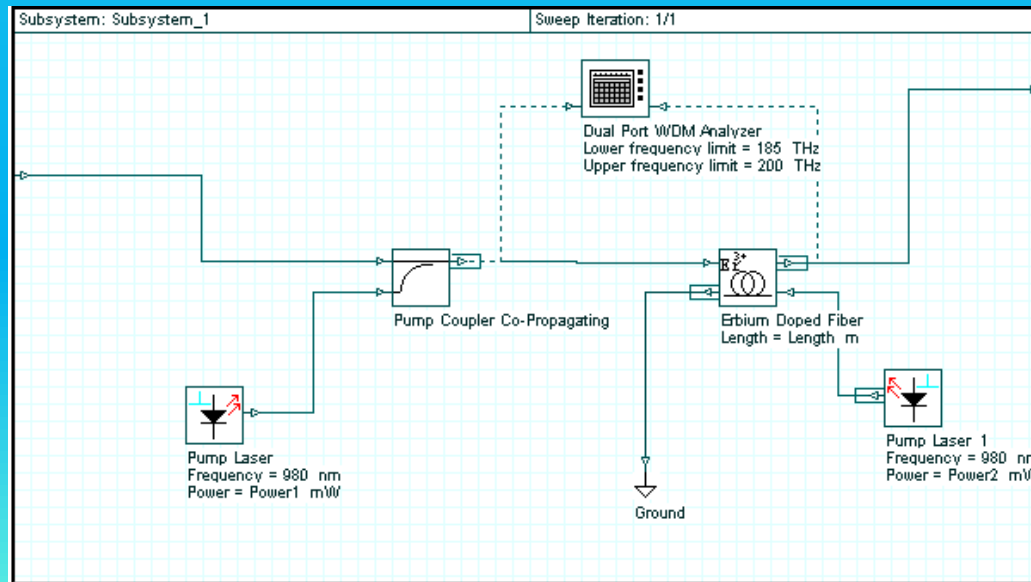


Oscilloscope





Dual Port WDM Analyzer



Calculated Results

Dual Port WDM Analyzer

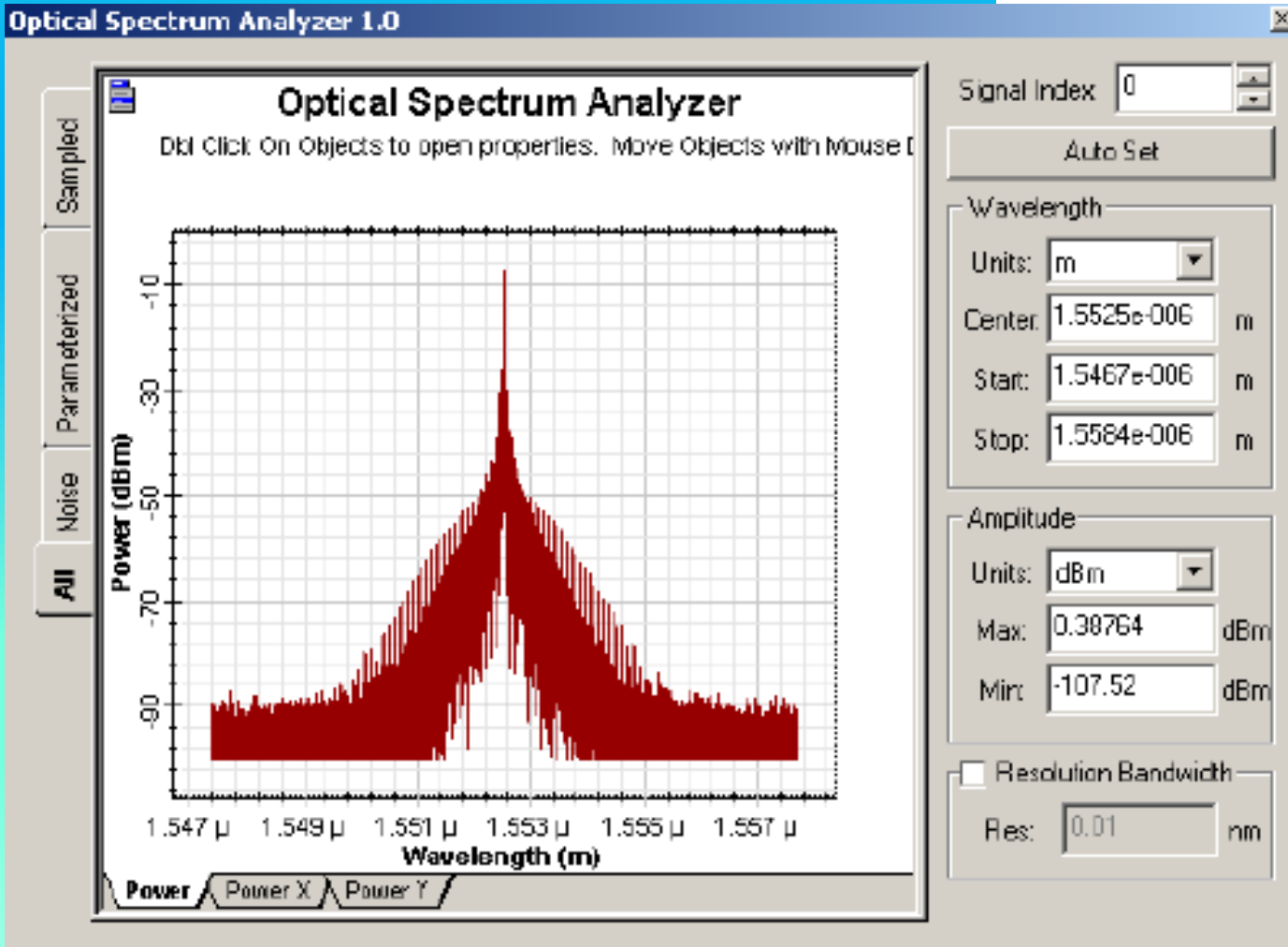
Frequency (THz)	Gain (dB)	Noise Figure (dB)	Input Signal (dBm)	Output Signal (dBm)	Output Noise (dBm)	Output OSNR (dB)
193.1	29.76227	4.21943	-30.3357	-0.57345	-23.9805	23.4071
193.3	30.017895	4.1469	-30.5583	-0.5404	-23.7929	23.2525
193.5	30.192835	4.30219	-31.095	-0.902136	-23.458	22.5559
193.7	30.416544	4.35884	-31.6104	-1.19383	-23.1731	21.9793

Analysis / Details



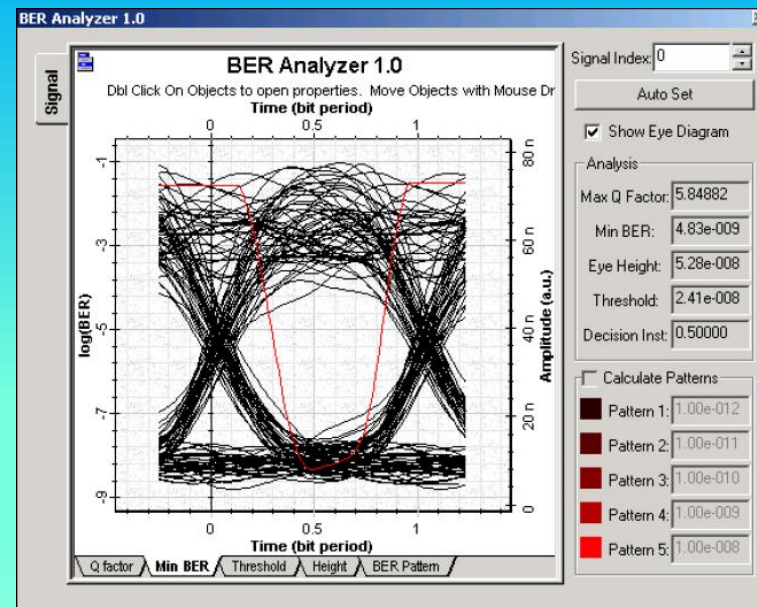
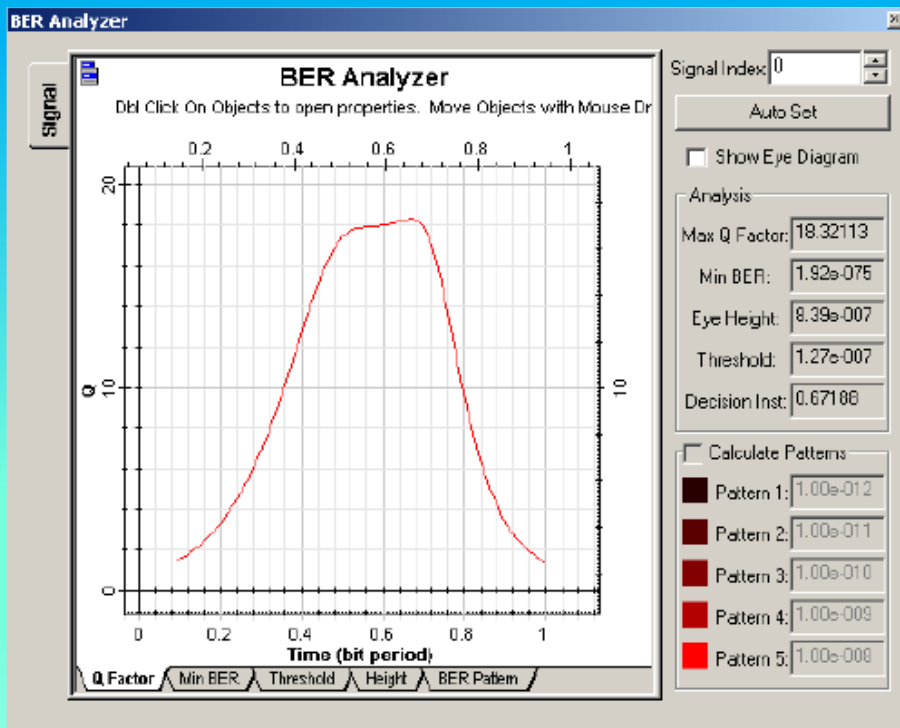
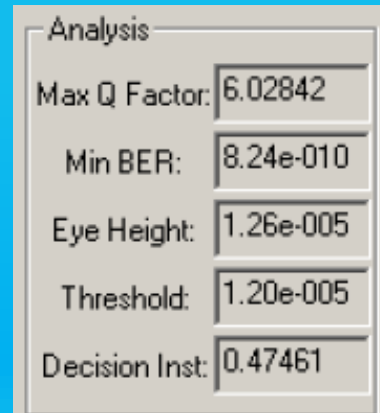
Spectrum Analyzer

Optical and Electrical





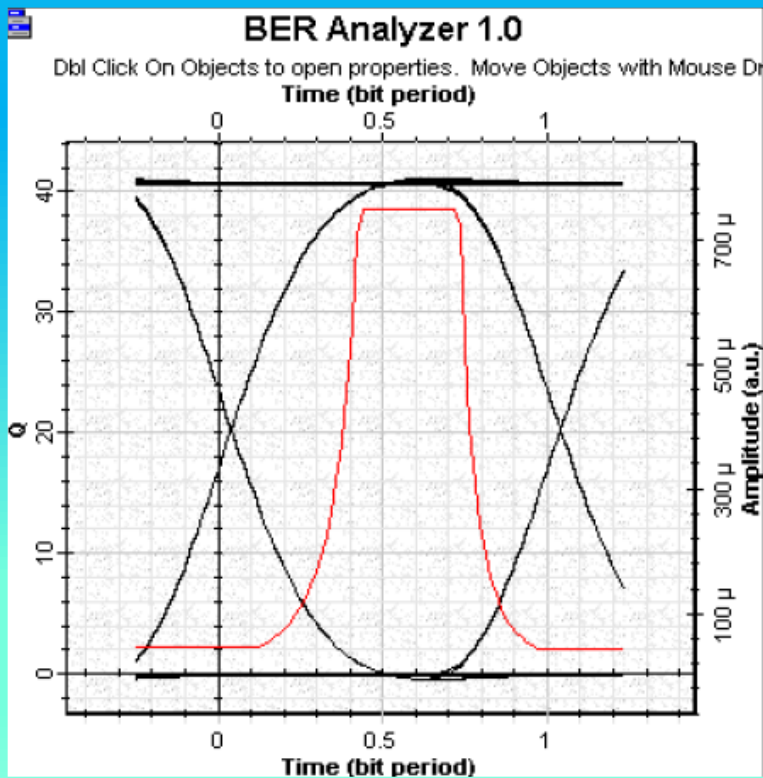
Eye Diagrama and BER



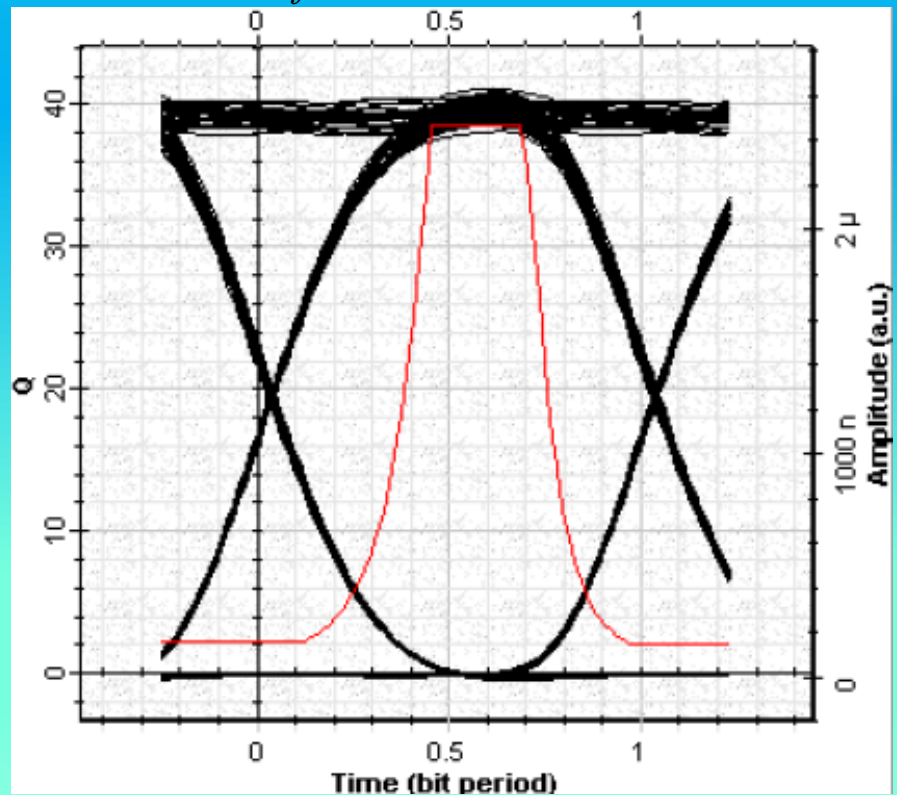


Q-Factor

$$L_{\text{fiber}} = 0 \text{ km}$$

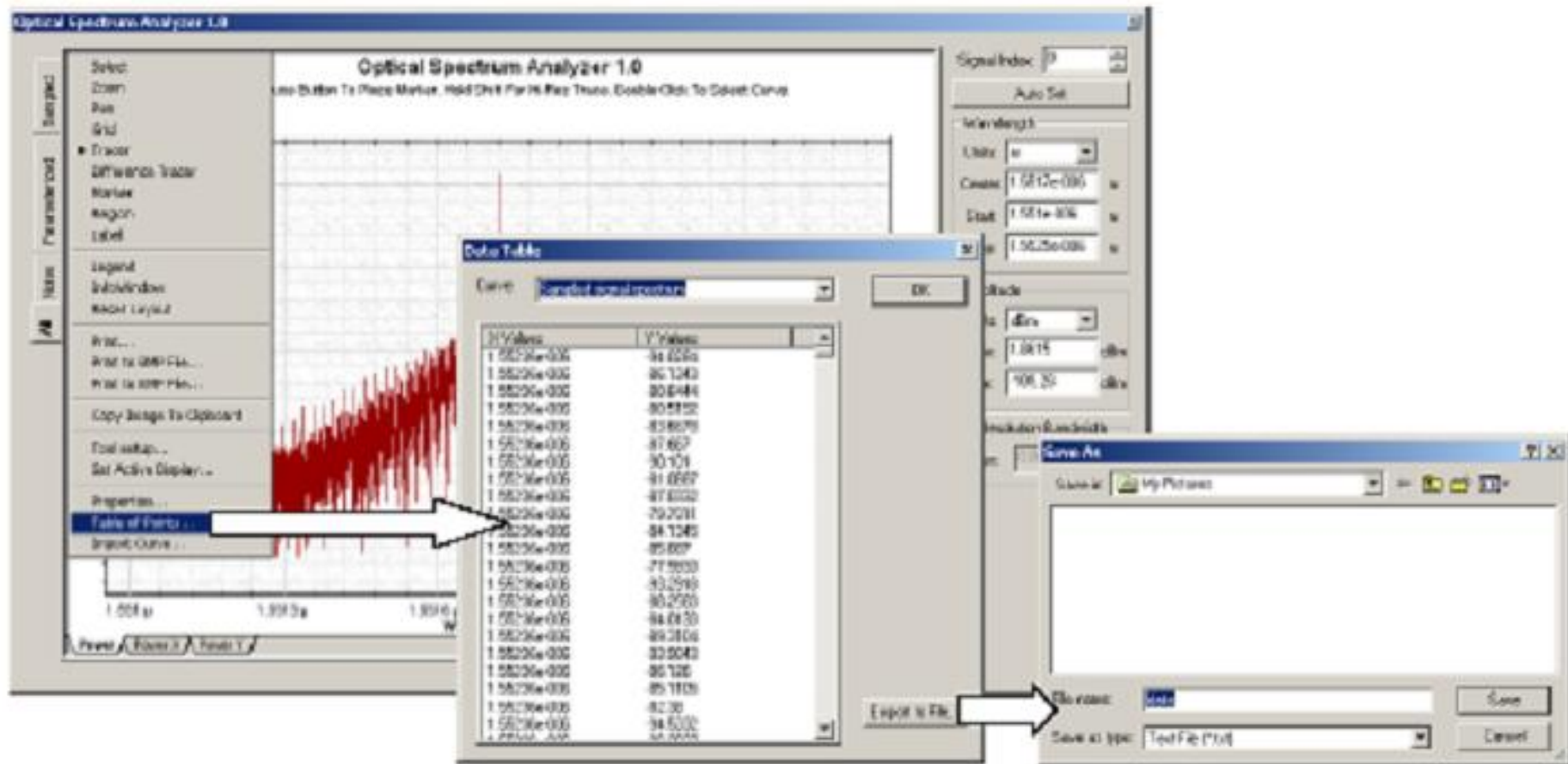


$$L_{\text{fiber}} = 100 \text{ km}$$



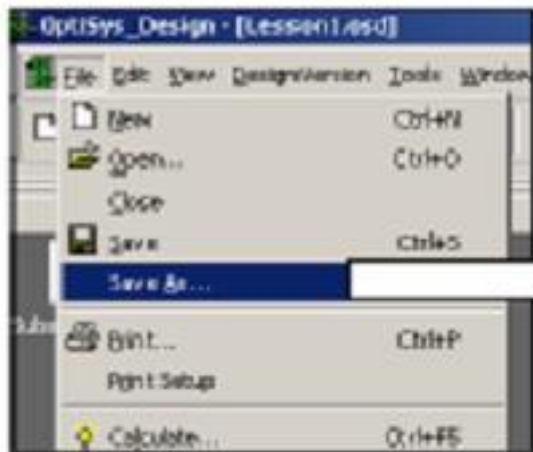


Exporting Calculated Curves

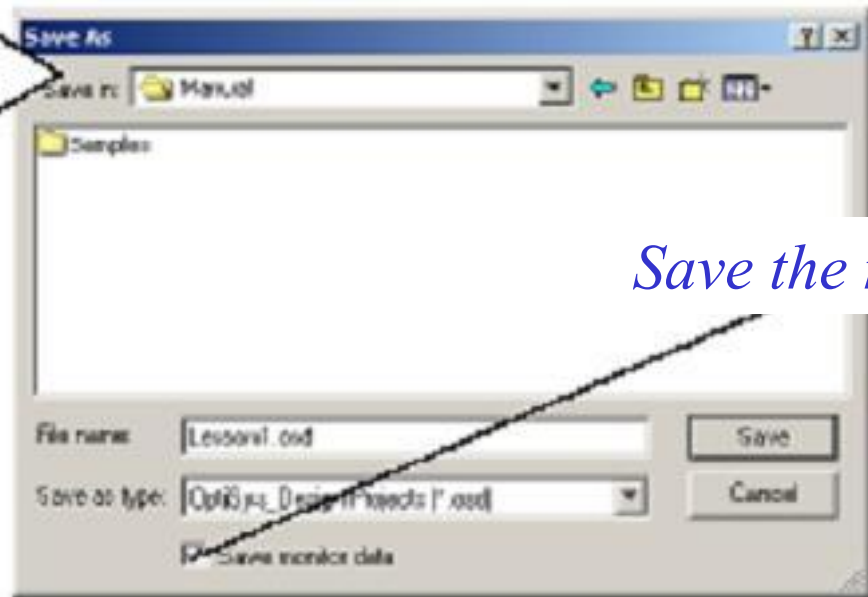




Saving Simulated Results



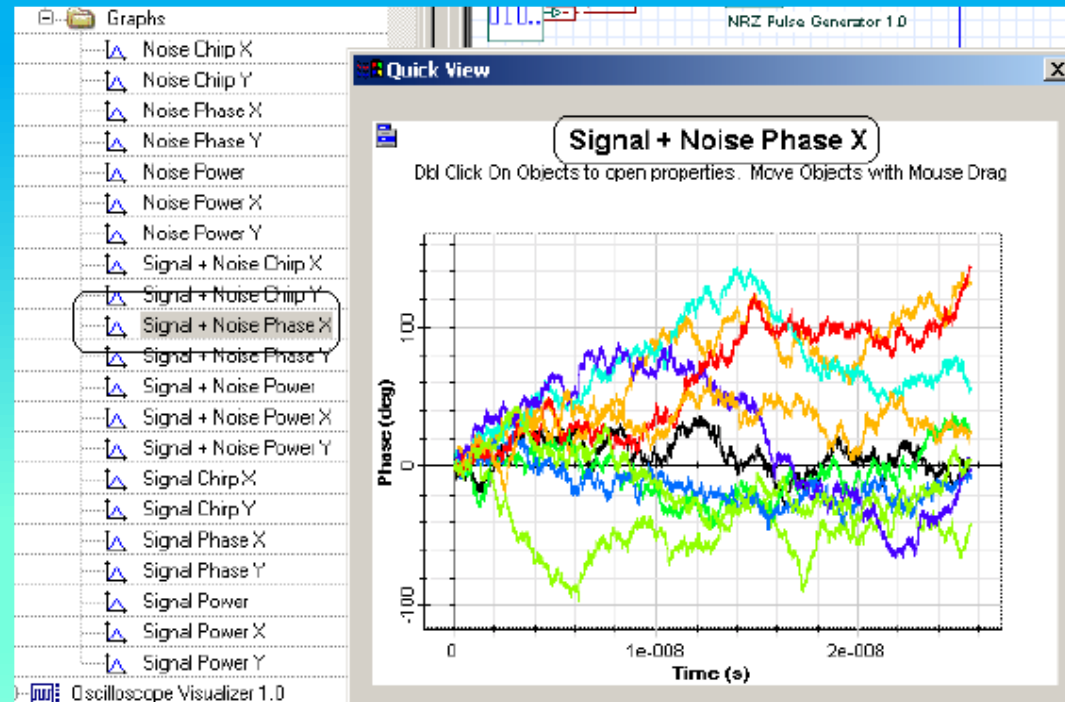
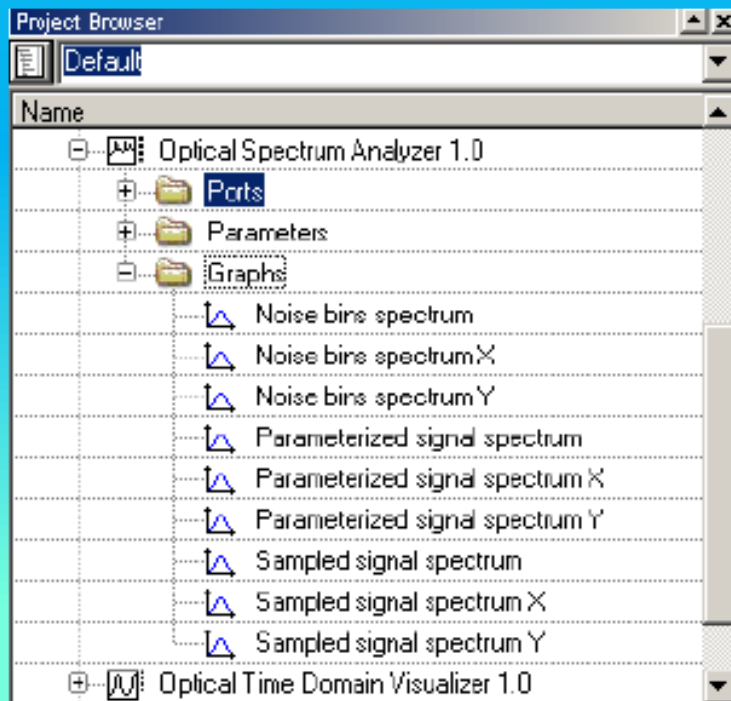
Requer mais memória



Save the results

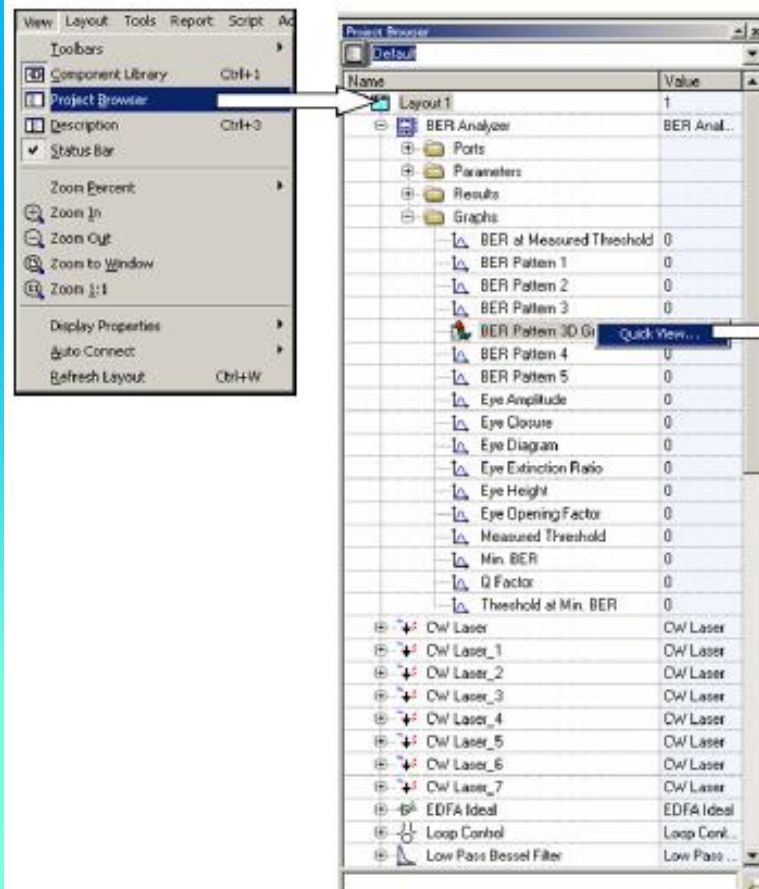


Additional Graphs at *Project Browser*

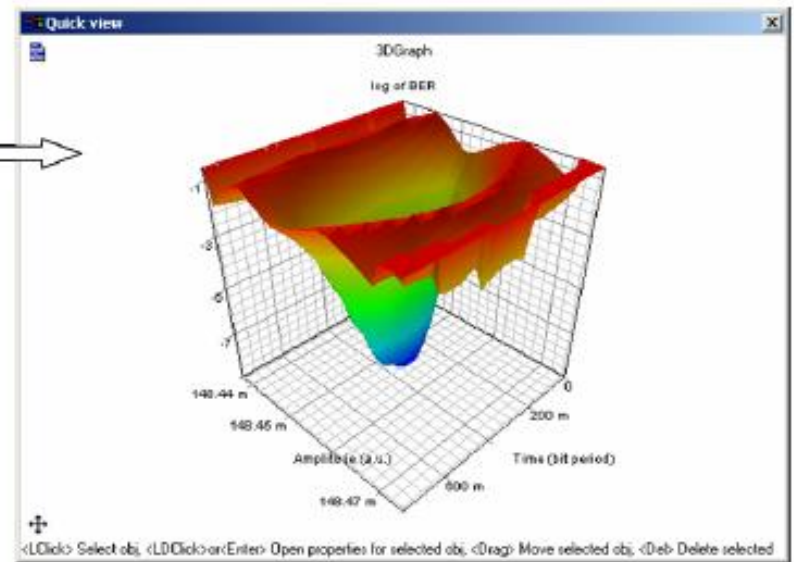




Graphs 3D

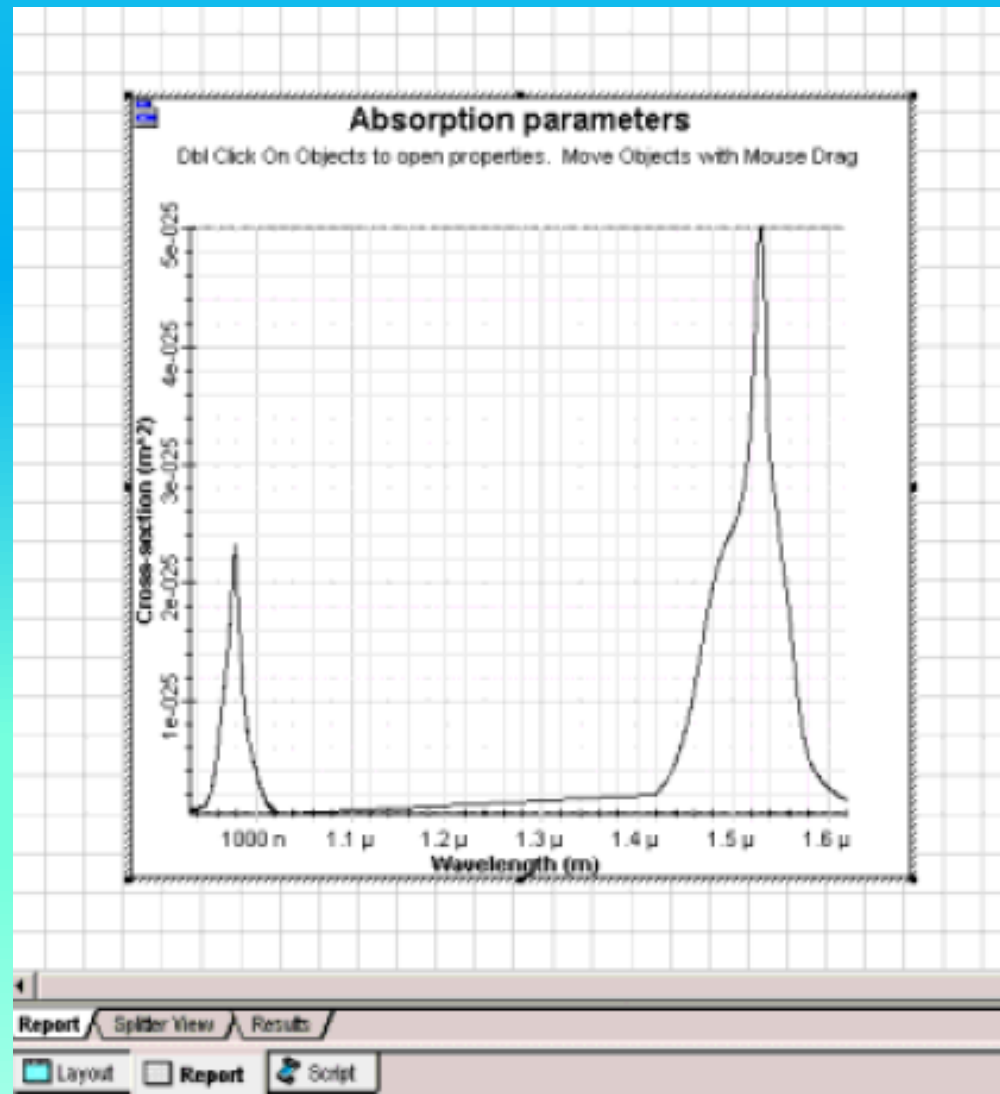


*Habilitar em Component
Propertiers*



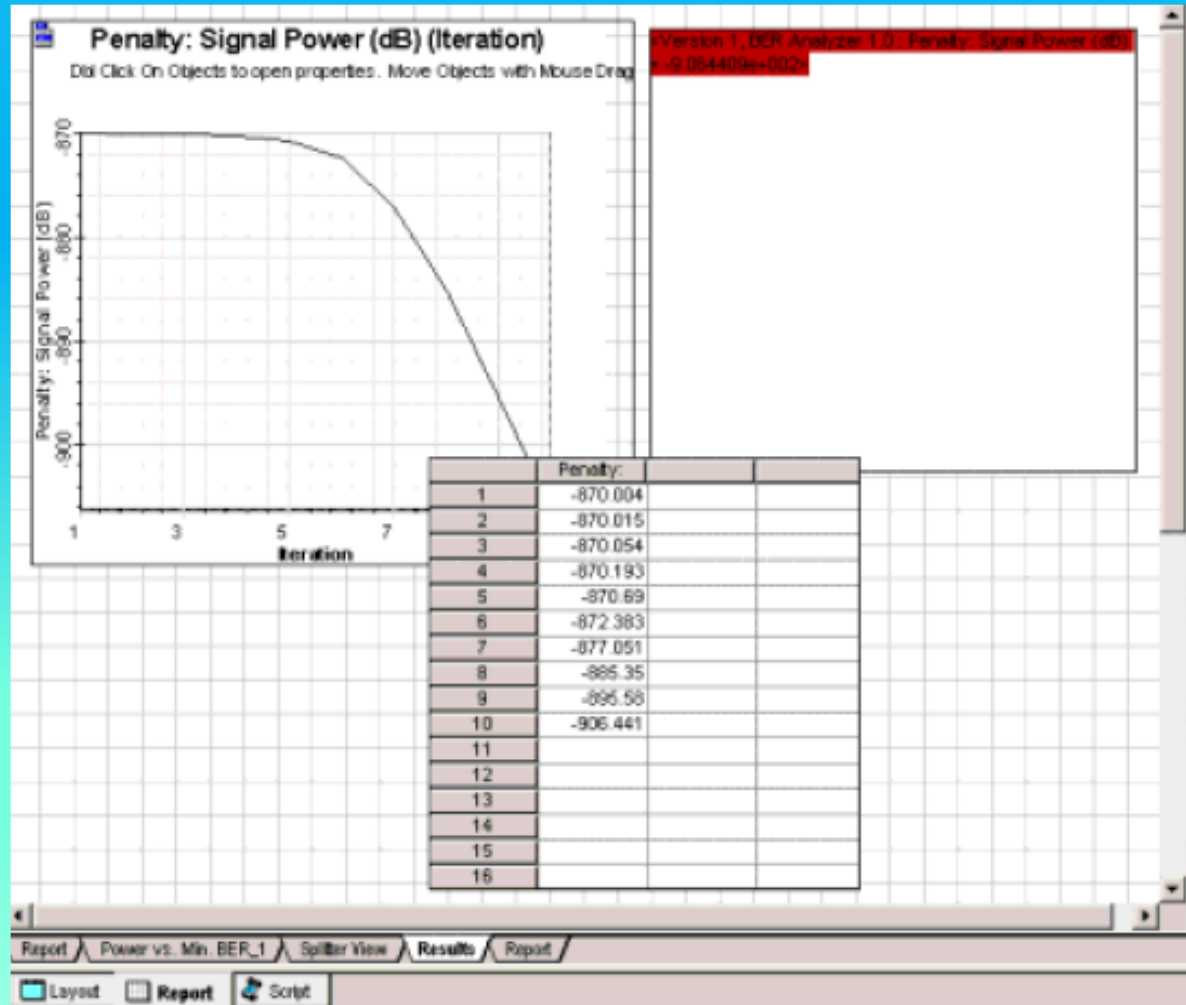


Building Graphs Using the *Report Window*



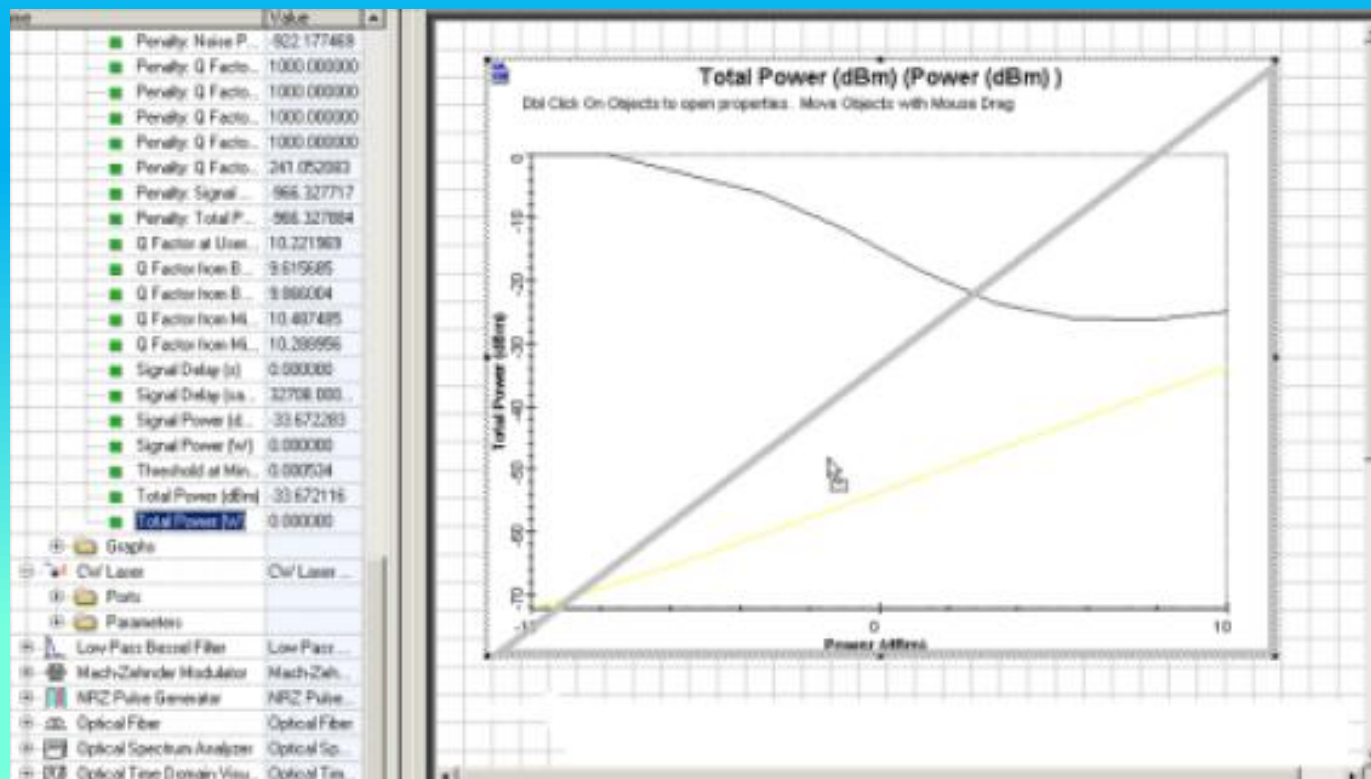


Resources to the Results' Analysis





Plotting Parameters with Results





Graphs

